

Participatory System Dynamics Modelling of Measles Vaccination, Transmission and Outbreak Response in Birmingham and Solihull

Dr David Ellis

Abstract

Background: Measles outbreaks are becoming increasingly common in the UK coinciding with declining MMR vaccination coverage, particularly in Birmingham and Solihull (BSol). Geographic, ethnic, and socioeconomic inequalities in vaccination coverage result in substantial variation in vulnerability to measles infection.

Aims: This study aims to investigate how a stakeholder-informed system dynamics model can support the development of interventions to increase MMR vaccination coverage and reduce outbreak burden.

Methods: A rapid literature review investigated barriers to childhood MMR vaccination. A stakeholder workshop with local public health and NHS professionals elicited views on barriers, facilitators, and potential interventions. A system dynamics model incorporating five population states across three age groups was then developed and calibrated before exploring the impact of varying childhood vaccination rates and different outbreak-management scenarios on outbreak burden.

Results: The rapid literature review identified vaccine accessibility, information/health literacy, trust, and other vaccine concerns as recurring factors preventing vaccination. Stakeholders identified vaccine accessibility and health literacy as the highest-priority factors, and prioritised extending vaccination into community settings and targeted communications as interventions for further investigation. The calibrated model reproduced the observed epidemic curve and hospital admissions, with modelled totals reaching 92% and 98% of observed values, respectively. Simulations estimated that reducing the delay to isolation substantially reduced outbreak burden, whereas increasing the isolation rate had a smaller effect. Furthermore, reductions in annual MMR vaccination produced disproportionately larger increases in infections and hospitalisations, particularly when sustained over longer periods.

Conclusion: The findings demonstrate the potential of a stakeholder-informed system dynamics model to support local measles elimination planning. While further development and validation are needed before the model can be used to evaluate specific interventions, the present model represents a substantial first step towards the development of such a decision-support tool.

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Term	Definition
Basic reproduction number (R_0)	The average number of secondary infections generated by one infectious individual in a wholly susceptible population.
Calibration	The process of adjusting model parameters so that model outputs reproduce observed data.
Herd immunity	A state in which a sufficiently large proportion of a population is immune that sustained transmission of an infectious disease is prevented.
Homogeneous mixing	An assumption that individuals in a population have an equal probability of coming into contact with one another, irrespective of characteristics such as age or ethnicity.
Incidence	The occurrence of new cases of a disease within a population over a specified period.
Susceptibility	The likelihood that an individual will become infected following exposure to an infectious agent.
Susceptible population	Individuals who are not immune to measles and are therefore capable of becoming infected if exposed to the virus.
Transmission	The process by which an infectious agent is passed from an infected individual to a susceptible individual.
Vaccine coverage	Proportion of a target population that has received vaccination.
Vaccine hesitancy	Delay in acceptance or refusal of vaccination despite the availability of vaccination services.

Acronym	Meaning
BCC	Birmingham City Council
BSol	Birmingham and Solihull
ED	Emergency department
FDP	Federated Data Platform
ICB	Integrated Care Board
IMD	Index of Multiple Deprivation
JCVI	Joint Committee on Vaccination and Immunisation
MMR	Measles, mumps and rubella
MMRV	Measles, mumps, rubella and varicella
NHSE	NHS England
NHS	National Health Service
ONS	Office for National Statistics
PHE	Public Health England
SIR	Susceptible–infectious–recovered
SMBC	Solihull Metropolitan Borough Council
UASC	Unaccompanied asylum-seeking children
UKHSA	UK Health Security Agency
WHO	World Health Organization

Student Contribution: I was responsible for the research idea, study design, recruitment of stakeholders to the collaborative modelling workshop, data collection, model development, statistical analysis, data visualisation, and the preparation and writing of the dissertation.

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1 Introduction

1.1 Measles Epidemiology

Measles is a highly transmissible viral disease that can cause serious complications and death [1]. Measles is primarily transmitted through the respiratory route, including airborne transmission from breathing contaminated air. Challenges in controlling airborne respiratory infections were highlighted during the COVID-19 pandemic [2]. As with COVID-19, transmission before the onset of symptoms presents challenges for controlling measles. For measles, infected persons are individuals typically infectious from around four days before until four days after rash onset [1]. Additionally, with a basic reproduction number R_0 often cited between 12 and 18, measles is considerably more transmissible than most respiratory infections and therefore presents a substantial challenge to population control [3, 4].

A substantial proportion of those infected with measles will develop complications, some of which require hospital care. Potential complications from the disease include blindness, encephalitis, severe diarrhoea, ear infections, and breathing problems including pneumonia. Complications from measles infection can also be fatal with an estimated 95,000 people globally dying from measles in 2024. Most of these deaths occurred in low-income countries with weak health infrastructures [1]. However, measles complications still pose a significant challenge in high-income countries where complications occur in around 10-20% of cases, and the case-fatality ratio is around 0.05% [5, 6]. Clinically vulnerable groups, such as immunosuppressed individuals, infants under one year, and pregnant people, are more likely to develop complications [7].

Immunity to measles can be obtained either following infection or vaccination. Infants can also obtain temporary maternal immunity from their mothers [8]. Herd immunity is achieved when the proportion of the population immune to a disease is high enough to reduce transmission to a level at which sustained spread cannot be maintained. Assuming homogeneous mixing of completely immune individuals within a population, the critical proportion required for herd immunity is approximately $1 - 1/R_0$. Therefore, more infectious diseases require a higher proportion of the population to be immune.

The measles vaccine was first introduced in the UK in 1968 and later replaced by the combined measles, mumps, and rubella (MMR) vaccine in 1988 [9]. The MMR vaccine is highly effective, with an estimated effectiveness of 92% (95% Confidence interval [CI], 67%–98%) after one dose and 95% (95% CI, 82%–98%) after two doses [10, 11]. To interrupt measles transmission and maintain elimination, WHO recommends achieving and sustaining at least 95% coverage with two doses of measles-containing vaccine [12]. However, substantial heterogeneity in vaccine uptake, and hence infections, has been observed between population groups within the same country/region [13]. Therefore, achieving a high overall coverage may not be sufficient to suppress measles transmission and protect the whole population.

1.2 Measles in Birmingham and Solihull

Birmingham and Solihull (BSol) are two local authorities in the West Midlands that together form the geographical footprint of the Birmingham and Solihull Integrated Care Board (BSol ICB). In 2024, the ICB covered an estimated population of 1,404,860, with 6.2% aged under five compared with 5.2% nationally [14]. BSol is also characterised by high levels of ethnic diversity and socioeconomic deprivation. Data from the 2021 Census show that just under half (48.5%) of BSol residents identified as White British, compared with 73.5% nationally. The largest ethnic groups were Pakistani (14.9%), Indian (5.7%), Black African (5.0%), and Other White (3.8%) [15]. Socioeconomic deprivation is not experienced equally across the two local authorities. Using the Index of Multiple Deprivation 2025, Birmingham and Solihull ranked 8th and 204th respectively among English local authorities. Furthermore, 42.8% of neighbourhoods in Birmingham and 13.4% in Solihull were among the most deprived 10% nationally ¹ [16, 17].

Under the current UK vaccination schedule, all children are offered a first dose of MMR when they turn 12 months and a second dose at 18 months [18].² The UK achieved measles elimination in 2016, meaning that endemic transmission had been interrupted, although imported cases continued to occur [18]. However, measles cases and outbreaks have increased in the UK with substantial internal transmission, coinciding with declines in vaccination coverage [20]. Figure 1 shows that there has been a series of measles outbreaks in England since January 2012.

Figure 1 also shows that these outbreaks have occurred alongside falling national vaccination coverage from a peak of 92.7% in 2013 down to 88.9% in 2024. However, in BSol, the coverage started lower and has subsequently fallen further than the England average from a peak of 89.7% in 2014 down to 81.3% in 2024 [22]. This decline has left local and national coverage substantially below the 95% WHO target for maintaining measles elimination and has coincided with substantial measles outbreaks in BSol [23].

There is also substantial inequality in MMR vaccination coverage and therefore susceptibility to measles. Figure 2 shows that, in February 2026, with the exception of Indian and Chinese children, MMR coverage was lower across all reported ethnic groups than among White British children. Coverage among Black African and Black Caribbean children aged less than six years was particularly low at 74.2% and 64.0% respectively compared to 92.9% for White British [24]. Additionally, children under six years living in the most deprived decile had an MMR coverage of just 62.7%, which was significantly lower than that among children living in the least deprived decile at 77.6%.

During the outbreak in Birmingham from 13 October 2023 to 12 April 2024, the in-

¹Based on Lower Super Output Areas (LSOAs).

²From 1 January 2026, the varicella vaccine was added to the routine childhood immunisation programme as part of the combined MMRV vaccine [19]. For brevity, both MMR and MMRV are referred to as MMR throughout this dissertation.

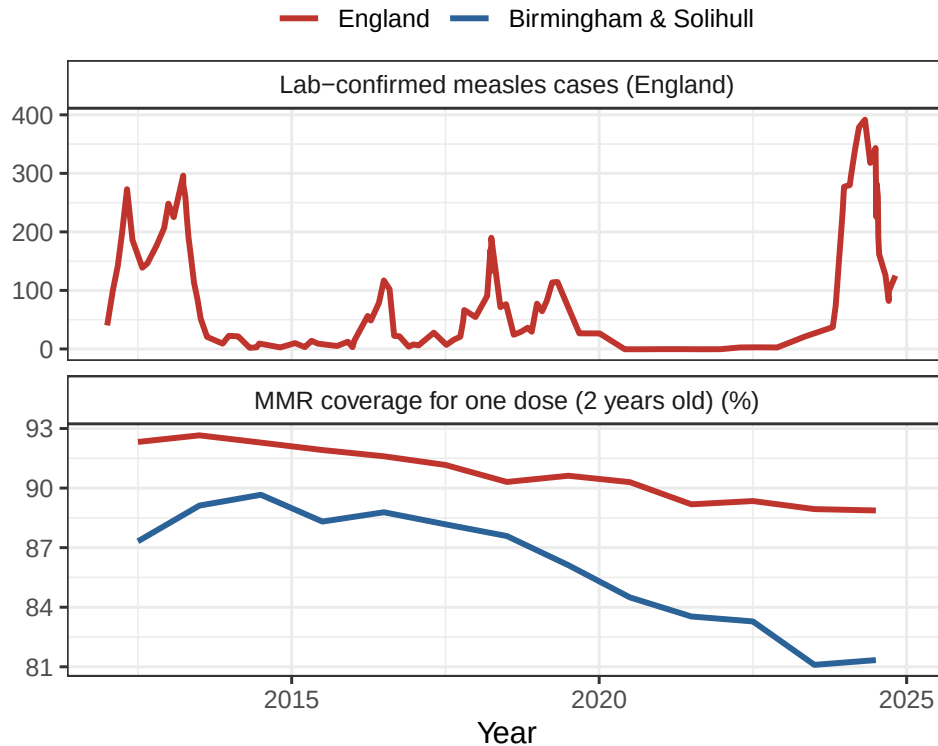


Figure 1: (Top) Monthly lab-confirmed cases of measles in England [21]. (Bottom) Yearly MMR coverage for one dose at two years old [22].

fection rate among Black African residents was eight times that among White British residents. Furthermore, the measles infection rate was 3.4 times that among those living in the third deprivation quintile. No infections were recorded amongst residents living in the least deprived quintile [23].

The factors influencing MMR vaccination uptake are complex, context-specific, and may vary between population groups, potentially contributing to inequalities in coverage [25–29]. Given the substantial ethnic and socioeconomic inequalities in MMR coverage observed in BSol, understanding these factors and how they vary between population groups is important for informing the development of the measles system model and may help to identify locally appropriate interventions to improve coverage and reduce inequalities in measles-related health outcomes.

1.3 Rationale and Aims

Parents' decisions about whether to vaccinate their children against measles are influenced by a complex combination of factors. Interactions between these factors, vaccination uptake, population immunity and measles transmission can produce complex dynamic behaviour, including feedback loops, non-linear relationships

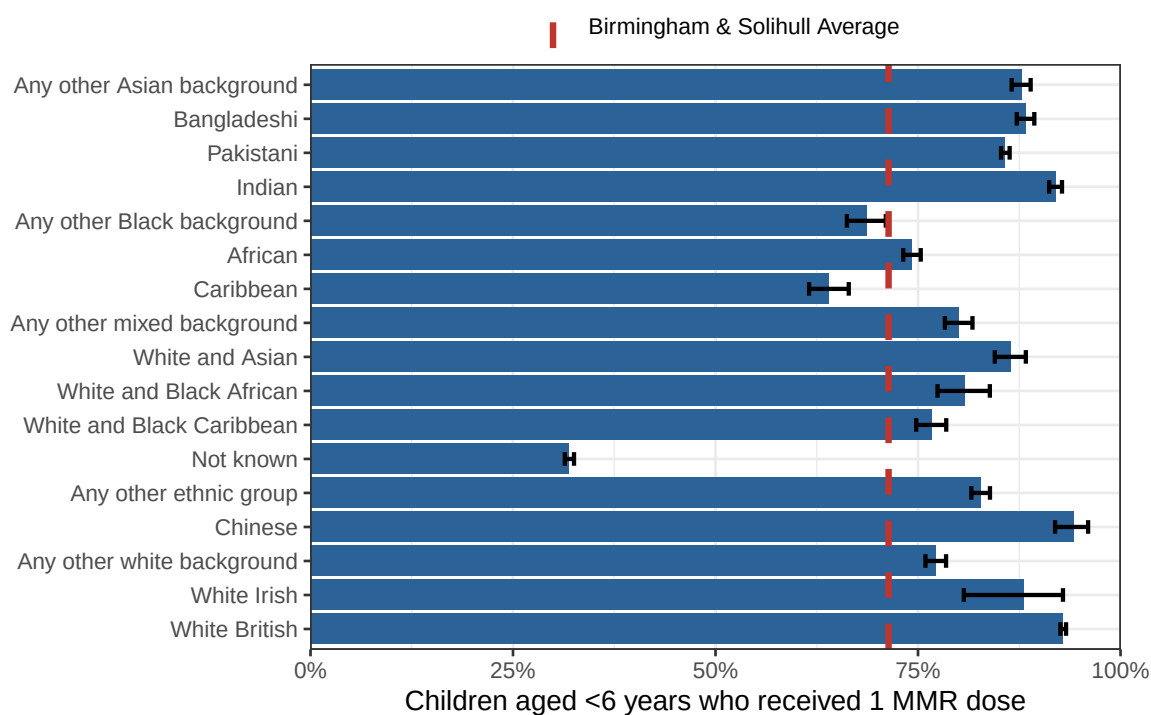


Figure 2: Percentage of children aged less than six years old living in BSol who have received at least one dose of MMR as of 04/02/2026 [24]. The error bars indicate the 95% CI calculated using the Wilson method.

and time delays. Simulating these interrelated factors therefore requires a modelling framework capable of representing interactions between behavioural and epidemiological processes over time.

System dynamics provides a framework for studying such complex systems and the interactions between their constituent elements [30]. Within this framework, systems are represented using stocks, which describe quantities that accumulate or deplete over time, and flows, which determine the rates at which these quantities change. In public health, stocks may represent populations of people, such as susceptible or immune individuals, but can also represent resources such as hospital beds or medicine supplies. System dynamics can also incorporate more abstract behavioural and social factors, such as vaccine confidence and health literacy, and represent their hypothesised relationships with vaccination behaviour [31]. System dynamics has been widely applied to health and medicine, including disease-related modelling, patient and workforce management, and healthcare system modelling [32–34]. The combination of epidemiological, behavioural and system-level processes considered in this study therefore makes system dynamics a suitable framework for modelling the interactions between vaccination uptake, measles transmission and disease burden in BSol.

This dissertation presents preliminary findings from an ongoing piece of work in

collaboration with stakeholders from Birmingham City Council (BCC), Solihull Metropolitan Borough Council, and BSol ICB. The findings from this project will contribute to future work packages for the Birmingham and Solihull Measles Elimination Group [35].

The overarching project aim is to develop a system dynamics model to support local decision-making through collaboration with a multidisciplinary group of stakeholders working across BSol. The intended full model will incorporate ethnicity-dependent behavioural and epidemiological factors, identified by stakeholders through a series of participatory modelling workshops, that facilitate and block flows in the system. The model will also simulate potential interventions intended to improve vaccination uptake and reduce measles transmission.

This dissertation first presents the findings of a rapid literature review on barriers to childhood MMR vaccination. It then presents findings from a participatory modelling workshop, including factors identified as influencing parents' decisions to vaccinate their children and interventions prioritised for further investigation as potential responses to declining MMR coverage in BSol. Lastly, the dissertation describes the development of a measles system model including transmission, vaccination and hospitalisation and presents preliminary predictions on the impact of modified outbreak preparation and response on the burden of measles outbreaks.

2 Methods

2.1 Literature Review

The search was designed to identify literature describing barriers and vaccine hesitancy among parents or carers in relation to childhood MMR vaccination. A semi-structured approach was adopted, with literature identified through searches of PubMed and ProQuest, supplemented by additional publications and grey literature identified through other sources. Only English-language publications were considered due to resource and translation constraints. To prioritise contemporary evidence while maintaining a manageable body of literature, only literature published in the last 10 years was considered.

The following literature search string was used to identify publications on barriers to childhood MMR vaccination experienced by parents and carers:

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(vaccine OR vaccination OR immunisation)
AND (barrier OR hesitancy)
AND (child OR parent)
AND (measles OR MMR)
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Titles and abstracts were screened for relevance, with potentially relevant publications reviewed in full. Findings were synthesised narratively to identify recurring

factors influencing vaccination decisions. A written summary of the findings was provided to stakeholders before the workshop to establish a shared evidence base, complement existing knowledge, and inform discussion of the factors and relationships to be represented in the measles system model.

2.2 Stakeholder Engagement

Due to the complicated and multifaceted nature of the measles system, including vaccination, disease transmission, healthcare provision, and interventions to improve vaccine uptake, stakeholders with expertise in different aspects of the system were invited to a participatory modelling workshop. Organisations were invited to nominate delegates to participate. These organisations included BSol ICB, the UK Health Security Agency (UKHSA), and the public health teams from BCC, and Solihull Metropolitan Borough Council. Attempts were also made to ensure that a range of skill sets were represented at the workshop including measles case management, child vaccinations, data analytics, and behavioural science.

To promote full participation, stakeholders were divided into two groups, with group composition selected to provide a broad range of backgrounds and expertise. A simplified version of the stakeholder engagement approach used by Michail et al. was employed, incorporating a simple system diagram prompt and a combination of small-group discussions, whole-group facilitated discussions, and sticky notes to elicit and capture stakeholder ideas and knowledge [36].

Prior to the workshop, participants were provided with an evidence pack³ via email outlining the rationale for using system dynamics and summarising local evidence on the systemic, sociological, and psychological barriers to childhood vaccination described in Section 3.1. The workshop was held at Birmingham City Council House and was facilitated with support from the BCC Governance Team. Participant contributions were recorded using paper and sticky notes, with all materials photographed and subsequently transcribed.

The workshop was divided into three parts. First, participants were given a high-level overview of the evidence pack and an introduction to system dynamics. A practical model-building exercise using flows of money into and out of a bank account was then used to introduce the core concepts of system dynamics.

Second, participants were presented with a simplified measles model showing the principal stocks and flows within the system. For each key flow, participants identified factors that could facilitate or block the flow using sticky notes. Participants were then given three stickers each and asked to affix their stickers next to factors they considered most influential using a limited number of stickers. Participants could allocate more than one sticker to the same factor, but the limited number of stickers required them to prioritise the factors they considered most important. This was used

³This evidence pack can be viewed on the project's [GitHub repository](#).

to focus subsequent model development on the factors considered most influential to system behaviour.

Finally, participants identified flows that they considered could be improved and proposed potential policies and interventions to alter these flows. For each intervention, participants identified the organisation responsible, the location within the system at which it would act, and the expected direction of its effect on the relevant flow. These contributions were also recorded using sticky notes. Participants subsequently used an anonymous online voting platform to rank the proposed interventions according to their perceived feasibility and potential impact. The resulting prioritisation was used to identify the interventions on which subsequent modelling would focus.

Working transparently with stakeholders is key to developing and maintaining trust and respect with the stakeholders and improves the overall legitimacy of the study [37]. Therefore, all materials from the workshop were digitally transcribed, results from participant voting were visualised and presented back to stakeholders for approval and validation at a later online meeting. Additionally, all group discussion activities were automatically transcribed using Microsoft Teams. Due to the workshop's length, Copilot was used to summarise this transcription. This summary was also presented back to the stakeholders to allow for accuracy checking.

2.3 Modelling

This study builds on the system dynamics model developed by Ackere and Schulz, which models measles vaccination, infection, and mortality for two age groups (children and adults) [31]. The present model retains these components while also modelling hospital admissions and the isolation of infected individuals during outbreaks, and disaggregates the population into three age groups (babies, children, and adults). A simple SIR model was first developed to establish the basic transmission dynamics and provide a foundation for subsequent extensions.

System dynamics modelling was conducted using Stella Architect [38], a proprietary software package with a graphical modelling interface. The visual representation facilitated engagement with stakeholders without technical expertise. Hospital admission data were extracted using SQL Server Management Studio (version 19.3.4). All data processing and visualisation were performed in R (version 4.3.3). All code and Stella Architect models used in this study are publicly available on [GitHub](#).

2.3.1 SIR model

To first simulate the most basic system behaviour, only three population states were included: susceptible (S), infectious (I), and recovered (R). It is assumed that these three populations are perfectly mixed and that everyone within the population has an equal probability of contacting each other person.

New infections occur at a rate determined by R_0 and the infectious period $1/\gamma$, while infected individuals recover at rate γ . For this simple model, the transition between states can be described by the following set of ordinary differential equations

$$\frac{dS}{dt} = -\frac{R_0\gamma SI}{N}, \quad \frac{dI}{dt} = \frac{R_0\gamma SI}{N} - \gamma I, \quad \frac{dR}{dt} = \gamma I. \quad (1)$$

Since this model does not include births or deaths, $N = S + I + R$ is a fixed value. The values for R_0 , γ , and N are taken from the literature and are presented in Table 5. The proportion of the population initially susceptible to measles, S/N , was taken to be 20%. This basic SIR model is shown by the blue components of Figure 4.

However, the basic SIR model does not account for the isolation of individuals following identification of infection. This is an important omission when modelling measles outbreaks, as the identification and isolation of infectious individuals is a key component of outbreak management. An isolation mechanism was therefore incorporated into the extended model to allow the potential impact of outbreak response measures on transmission to be investigated.

One simple way to model this is to implement an “isolation mode” that turns on a fixed number of days after the first measles case. Once this mode is activated, a daily fixed proportion of infected persons are moved to an isolated state in which they can no longer infect others. The time delay to response p_D and the isolation proportion p_P are free model parameters. Their values will be calibrated to observed data as outlined in Section 2.3.4. When the “isolation mode” is activated, the rate of change of infected persons is modified to be

$$\frac{dI}{dt} \rightarrow \frac{R_0\gamma SI}{N} - \gamma I - p_P I$$

Standard practice for communicable diseases is to only consider an outbreak “over” after two maximum incubation periods occur without any new cases [10]. Since the incubation period for measles is 7 to 18 days, the outbreak should be considered ongoing until 36 days have passed without a new case. After this point, authorities can stop looking for infected individuals.

This can be modelled with an “isolation switch”. The mechanics of this switch are demonstrated in Figure 3. If there is at least one person currently infected, and the switch isn’t currently on, then after a fixed delay p_D the switch turns “on” and a fixed proportion of infected persons are isolated each day. However, if there are no infected persons ($I(t) < 1$), then the switch will turn off after two maximum incubation periods (36 days).

Because the model represents population stocks as continuous quantities, an infected stock between zero and one would otherwise permit a fractional infectious

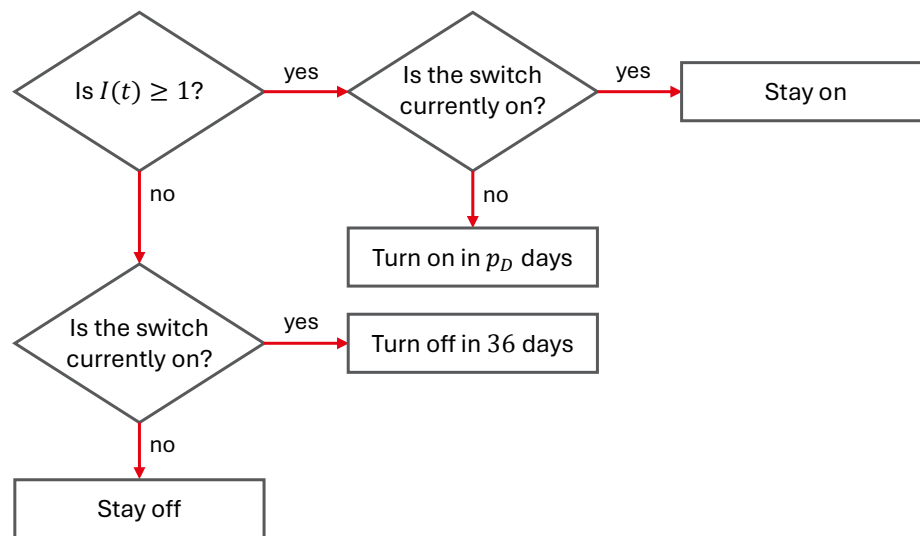


Figure 3: Flow diagram outlining the mechanics of the measles isolation switch.

population to generate further infections. To prevent this artefact, transmission was set to zero when the infected population fell below one individual.

The additional simulation components related to this isolation mechanism are presented in green in Figure 4.

2.3.2 Base Model

The base model, shown in Figure 5, is an extension of the SIR model with isolation to include three different age groups: babies (< one year), children (1 to 4 years), and "adults" (5+ years), as well as an additional hospitalised population state. This gives a total of 15 different population states (3 age groups $\times$ 5 states). In this model, hospitalised individuals are assumed to also be isolated. Therefore, the addition of this state will change the isolation delay (p_D) and proportion (p_P) needed to fit the model to observational data. The inclusion of a hospitalised state gives the model another output that can be compared and fitted to observed data. It is therefore necessary to include an additional free parameter p_H which controls the rate at which infected individuals are admitted to hospital.

Importantly, the base model also includes childhood vaccinations. A simplifying assumption was made that those who receive a single dose of MMR become fully protected from measles infection. This assumption is justified since MMR is estimated to 92% effective after just one dose [11]. This vaccination is therefore modelled as movement into the vaccinated/recovered state, since vaccinated individuals are assumed to have equivalent protection against measles infection to recovered individuals. Since only children aged one to four years old are routinely vaccinated, only this group includes a path between the susceptible and recovered state in the

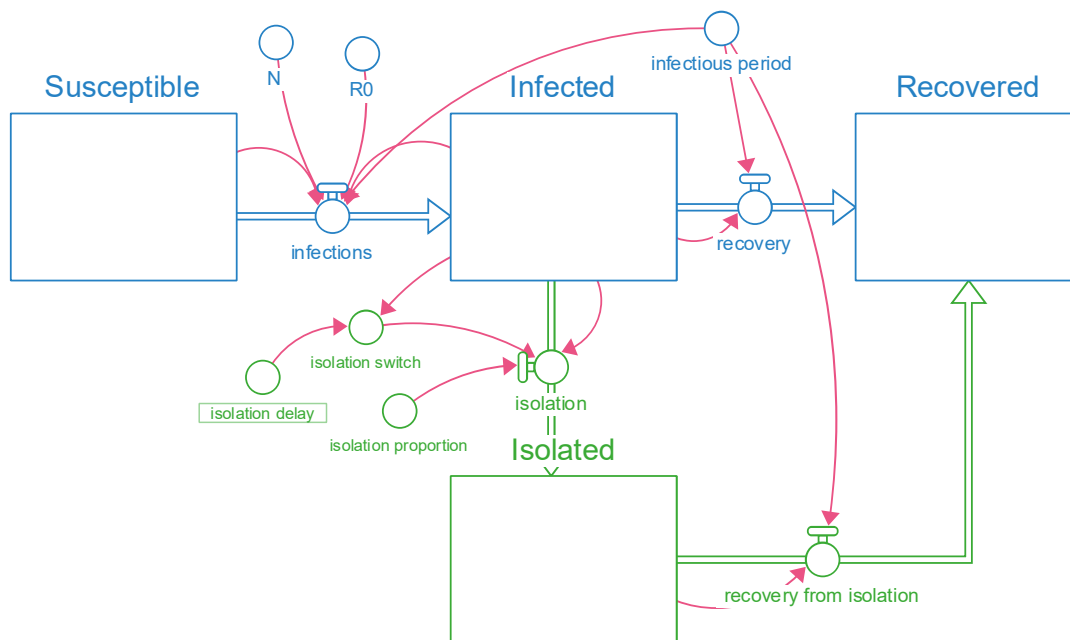


Figure 4: Diagram of a basic SIR system (blue) extended to include the ability to isolate infected individuals (green) at a fixed daily proportion p_P . This isolation mode is activated p_D days after the first infection has occurred.

base model. This number is taken to be the average number of vaccines given to children aged 1 to 4 years old in BSol between 2021 and 2025 [24].

The model also includes age-specific migration into Birmingham with vaccination proportions estimated by combining migration estimates with WHO vaccination estimates as outlined in Appendix 5 [15, 39, 40]. However, importantly, the model does not account for migration out of BSol or internal migration to and from other areas within the UK. Finally, the model includes age-specific all-cause mortality calculated using data from the deaths register for deaths recorded between 2022 and 2025 [41]. This date range was chosen to avoid the impact of the Covid-19 pandemic on the mortality estimates.

2.3.3 Age-specific rates

The probability of infection and the probability of hospital admission when infected are age-dependent [23, 31]. Following the approach of Ackere and Schulz, the basic reproduction number can be written as

$$R_0\gamma = \frac{\alpha_A P_A + \alpha_B P_B + \alpha_C P_C}{P_A + P_B + P_C}, \quad (3)$$

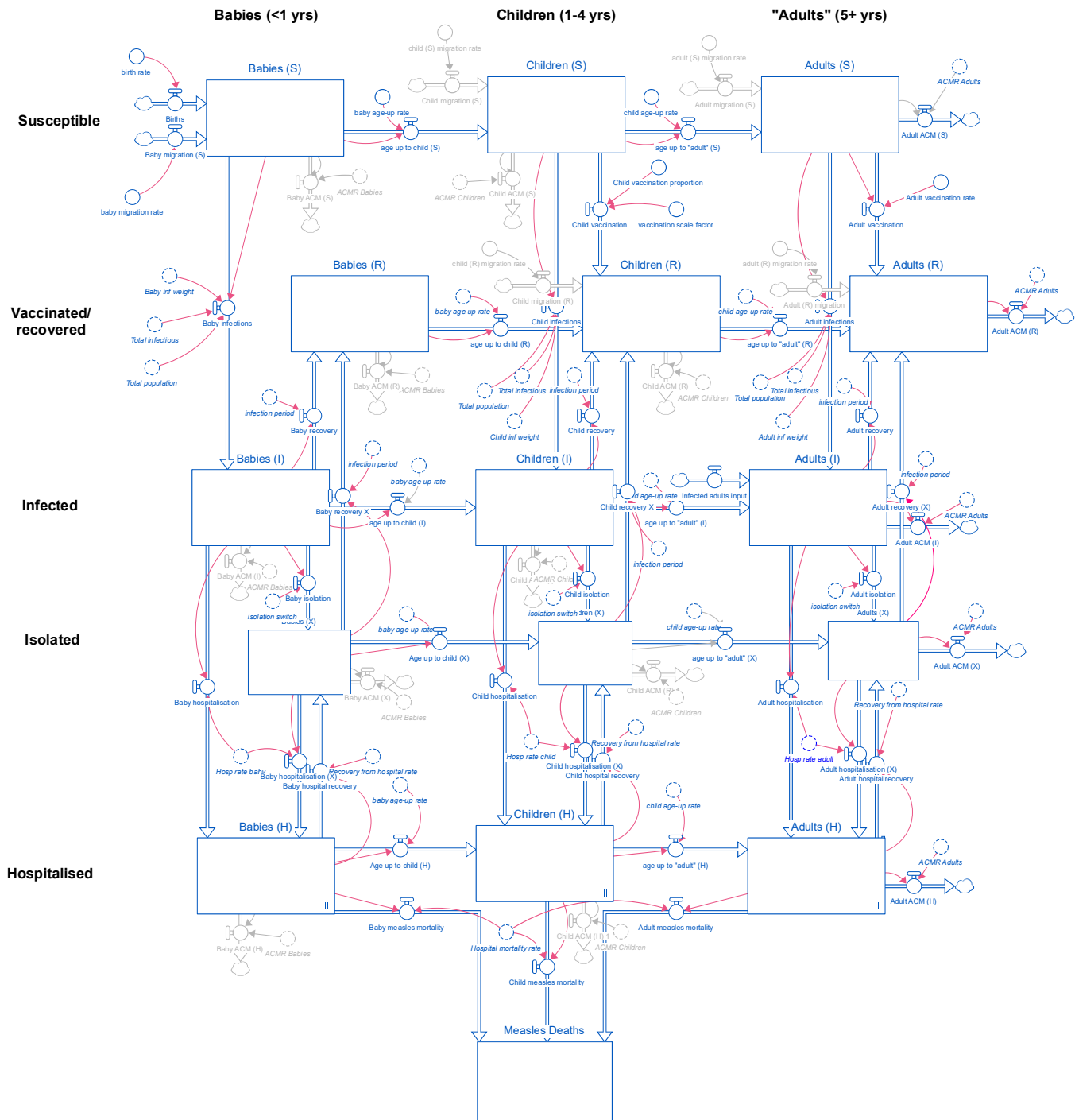


Figure 5: Base measles system model with five population states across three age groups. Boxes represent population stocks, taps represent population flows, blue and gray circles show model variables, and pink and gray arrows indicate information connections.

where α_i and P_i are the relative infection probability and population size respectively for adults (A), babies (B), and children (C) [31]. These parameters should be interpreted as effective age-specific transmission and hospital admission rates incorporating differences in exposure, susceptibility, and vulnerability rather than as estimates of biological susceptibility alone.

Age Group	Population	Cases	Hospital Admissions	Unvaccinated %	Infected per 1000 unvaccinated
<1	14,272	56	30	100	3.924
1-4	60,751	124	33	21.1	9.674
5+	1,069,893	226	64	10.7	1.974

Table 1: Number of infections per 1000 unvaccinated persons (Birmingham only). Populations from 2021 ONS census [15], age-specific cases from Jary et al., and vaccination coverage from NHS FDP [24] and WHO Immunisation Portal [40].

The number of infections per 1000 persons not vaccinated with at least one dose of MMR are presented in Table 1 for each age group. Assuming that observed infections per unvaccinated individual are proportional to the relative age-specific infection rates, the ratios of these rates can be estimated from the values in Table 1. It can hence be calculated that

$$\frac{\alpha_B}{\alpha_A} = 1.99, \quad \frac{\alpha_C}{\alpha_A} = 4.90. \quad (4)$$

Substituting into Equation 3, it can be therefore be shown that $\alpha_A = 1.27$, $\alpha_B = 2.52$, and $\alpha_C = 6.20$. The age-specific infection rates can then be calculated by making the replacement $R_0\gamma \rightarrow \alpha_i$.

Similarly, the observed ratio of hospital admissions among infected individuals (Table 1) was used to estimate relative age-specific hospitalisation rates

$$h_B = 1.89h_A, \quad h_C = 0.940h_A. \quad (5)$$

However, to ensure that the modelled hospital admissions match the observed numbers, the infected adult daily hospitalisation proportion h_A is left as a free parameter to be fitted.

The error function for weekly incidence E_I and hospital admissions E_H were then calculated using Equation 6 for a range of values for the isolation delay p_D , isolation proportion p_P , and adult hospitalisation proportion p_H . From this, the combination of parameters that minimises the combined error function was calculated.

2.3.4 Model Calibration

The model was calibrated using a grid-search approach, in which simulations were run for every combination of the free parameters across predefined parameter ranges. The daily model outputs were aggregated into weekly values and compared with observations from the large 2023/24 measles outbreak in BSol. Weekly laboratory-confirmed case data for Birmingham were extracted from Jary et al. [23]. Weekly data were not available for Solihull. A first-order approximation was therefore constructed by allocating the annual number of Solihull cases across weeks in proportion to the observed Birmingham epidemic curve (see Appendix 5). Measles-related hospital admissions data were calculated using Secondary User Statistics (SUS) from the Birmingham and Solihull Integrated Care Board (BSol ICB) data warehouse. All dominant hospital episodes with an ICD-10 code starting with 'B05' (measles) were included.

Both models were initialised with a single infectious individual to represent the introduction of measles into an otherwise susceptible population. For the base model, this individual was assigned to the adult age group. For simplicity, the remaining population in the SIR model was initiated such that $S/N = 0.9$. For the base model, the initial number of people in hospital or isolation was set to zero, and the number of people initially in the susceptible and recovered/vaccinated states were determined based on ONS census, NHS vaccination, and WHO vaccination data [15, 24, 40]. The exact initial conditions used are presented in Appendix 5.

To maximise the use of data, the model is calibrated by minimising an error function. As the objective is to reproduce the outbreak dynamics rather than the calendar timing of the outbreak, a temporal alignment parameter, τ , was applied during error calculation to allow the simulated epidemic to be shifted relative to the observed epidemic. For each parameter combination, the value of τ that minimised the error was selected. The error function is then given by

$$E_V(\mathbf{p}, \tau) = \sum_t (V_{t+\tau}^{\text{sim}}(\mathbf{p}) - V_t^{\text{obs}})^2, \quad (6)$$

where $\mathbf{p}$ represents all free model parameters to be tuned, and $V_{t+\tau}^{\text{sim}}$ and V_t^{obs} are the simulated and observed outcomes values, respectively.

There are two observable values for the model to be calibrated to: measles infections and hospital admissions. Therefore, the full error function to be minimised is

$$E(\mathbf{p}) = \frac{E_I(\mathbf{p})}{\text{SD}(E_I)} + \omega \frac{E_H(\mathbf{p})}{\text{SD}(E_H)}, \quad (7)$$

where E_I and E_H are the error functions for infections and hospital admissions, respectively and ω determines the relative weighting assigned to hospital admission error E_H compared with infection error E_I . The error functions are normalised by dividing by their standard deviation across all parameter combinations tested in the grid search. Since this investigation is most interested in measles infections, a value

of $\omega = 0.5$ was chosen. It should be noted that the choice for ω is subjective and that there is no “correct” value. The impact of the choice of ω on the fitted parameter values was investigated through sensitivity analysis. All model parameters used are presented in Table 5 of Appendix 5.

3 Results

3.1 Literature Review

There were 16 and 35 publications identified by PubMed and ProQuest, respectively. Five additional publications were included, four of which were grey literature based on local community engagement workshops conducted by BCC and Solihull Metropolitan Borough Council [42–45]. The publications included are presented in Table 4. Due to their increased relevance to the present work, the following review focuses particularly on the local evidence.

Evidence regarding the socioeconomic determinants of childhood MMR uptake in England is relatively limited and often based on older data. A 2017 systematic review of socioeconomic differences in childhood vaccinations found MMR uptake in England to be lower amongst parents with a higher “socioeconomic status” (SES) [46]. However, the evidence underlying this finding was limited and relatively old, with the most recent contributing study published in 2011, and results varied according to how socioeconomic status was measured [47–51]. In contrast, ONS data shows that MMR uptake is lowest in the most deprived local authorities based on the Index of Multiple Deprivation (IMD) [17, 22]. These more recent population-level data likely provide a better indication of the socioeconomic trends in MMR uptake in England.

Local community engagement undertaken in Birmingham and Solihull with ethnic groups experiencing low vaccination uptake identified several overlapping themes [42, 44, 45]. Language barriers and limited literacy arose as a key issue. Many parents in BSol speak little English and therefore struggle to benefit from English-only flyers while others are unable to read in any language. This highlights the importance of using multiple media forms, including video, for outreach [42, 44]. Participants also reported that the materials used to promote vaccination lacked diversity and did not represent the communities they were from [44]. This finding mirrors that of a 2024 systematic review on the barriers and enablers to MMR uptake among disadvantaged, minority, or underserved populations [52].

Concerns about the ingredients and safety of the vaccine varied by community. For example, Muslim participants were particularly concerned about pork gelatine and the option of a pork-free vaccine [42, 44]. Globally, religiously motivated concerns about measles vaccination have included the use of animal-derived gelatine and, in some contexts, the use of human foetal cell lines in vaccine production [53].

Concerns about a possible link between MMR vaccination and autism were reported as important for some Muslim/Afro-Caribbean participants, although these

fears were less prominent for the Romani traveller community [42, 44]. This local diversity in opinion is important as a 2023 systematic review of studies conducted in the US found the risk of autism to be the most commonly cited concern about the vaccine [28]. These fears originate in the now-discredited and retracted claim linking MMR vaccination with autism [54–57].

Accessibility was also identified as a barrier, particularly among migrants and asylum seekers [45]. A BCC survey of support staff working with unaccompanied asylum-seeking children (UASC) in Birmingham found that UASC often arrive with missing vaccination records, and that there often isn't enough time within appointments to give the required immunisations [43]. Additionally, a 2025 review found that many migrants living in Europe don't know how to access vaccine services and don't know what they're eligible for, including fears of costs [58]. This is notable because all migrants in the UK, including those with No Recourse to Public Funds (NRPF), are eligible for free vaccinations on the NHS [59, 60]. The same review found that first-generation immigrants had low vaccination rates and high levels of unknown vaccination status [58]. A 2019 systematic review of parental attitudes and beliefs of the measles vaccine in Europe also found that parents from Gypsy, Roma, and Traveller (GRT) populations faced unique accessibility barriers due to their transient residency as well as regular discrimination and exclusion within the healthcare system [61].

Participants in the engagement workshops conducted by BCC also raised the issue of trust [44, 45]. Some participants reported a general mistrust of government sources, particularly if associated with the 'Home Office' and 'immigration'. A general preference was raised for information to come from healthcare professionals and local community/faith leaders. A lack of trust in the healthcare system has also been found to be a barrier to vaccination in the UK [27].

Finally, broader perceptions of vaccination may also influence uptake, including concerns about receiving multiple vaccines at a young age, low perceived benefits of vaccination, and previous adverse experiences [28, 62].

3.2 Participatory System Modelling

A participatory measles modelling workshop was held on the 20th of March 2026. This workshop, hosted by Birmingham City Council, brought together local experts to build a conceptual model of the local measles system, identify behavioural and epidemiological factors that facilitate or block flows in the system, and identify potential interventions to improve these flows. Attendees to the workshop included employees from Birmingham City Council (7), Birmingham and Solihull Integrated Care Board (2), and Solihull Metropolitan Council (1).

Due to the workshop's length, Copilot was used to extract audio contributions to supplement the written materials. The full output, along with all non-sensitive materials related to this project, can be viewed on [GitHub](#).

3.2.1 Barriers and Facilitators to Vaccination

Workshop participants were separated into two groups and given a basic flow diagram for the local measles system with five connected population states: susceptible, vaccinated, infected, hospitalised, and recovered. The groups were asked to identify all the various factors influencing the rate of flow of people from one state to another. These factors were added to the relevant flow. The resulting models have been digitally transcribed and are presented in Appendix 5 as Figures 12 and 13. Photographs of the models produced in the workshop can also be found on [GitHub](#).

Once participants had collated all of the factors they could think of, they were asked to vote on which factors they deemed most important. Each participant was given three stickers with which they could vote on factors. These votes are also transcribed in Figures 12 and 13. After the workshop, these factors were grouped together and counted. The total number of votes for each flow factor is shown in Figure 6.

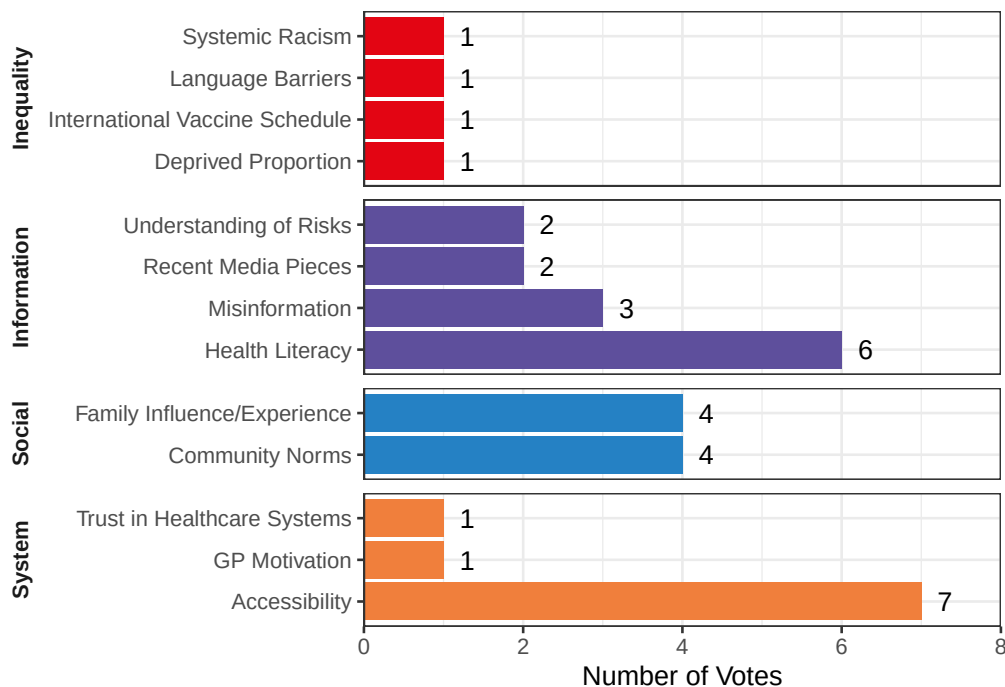


Figure 6: Number of participant votes on factors believed to play an important role in influencing parents' willingness to vaccinate their children against measles.

As shown in Figure 6, all of the factors voted on related to vaccination. The factor with the most votes was the accessibility of vaccination ($n = 7$), followed by health literacy ($n = 6$), the influence and previous experiences of family members ($n = 4$), and community norms ($n = 4$).

3.2.2 Prioritised Interventions

Participants were asked to review the models they had created and identify the “problematic” flows within the system. In their groups, they were asked to brainstorm a list of potential interventions to improve vaccine uptake and which organisations they thought would be responsible for these interventions. These interventions are also shown in Figures 12 and 13.

Each group was then asked to each choose the three interventions that they thought were worth prioritising for inclusion in the model. These six interventions were then anonymously ranked by each of the workshop attendees using Menti-meter. The results were then aggregated using a Borda count approach, in which each ranked position was assigned a decreasing number of points (six points for rank one, five points for rank two, etc) and the total score for each option was used to generate the final ordering [63, 64]. The total number of points for each intervention is shown in Figure 7.

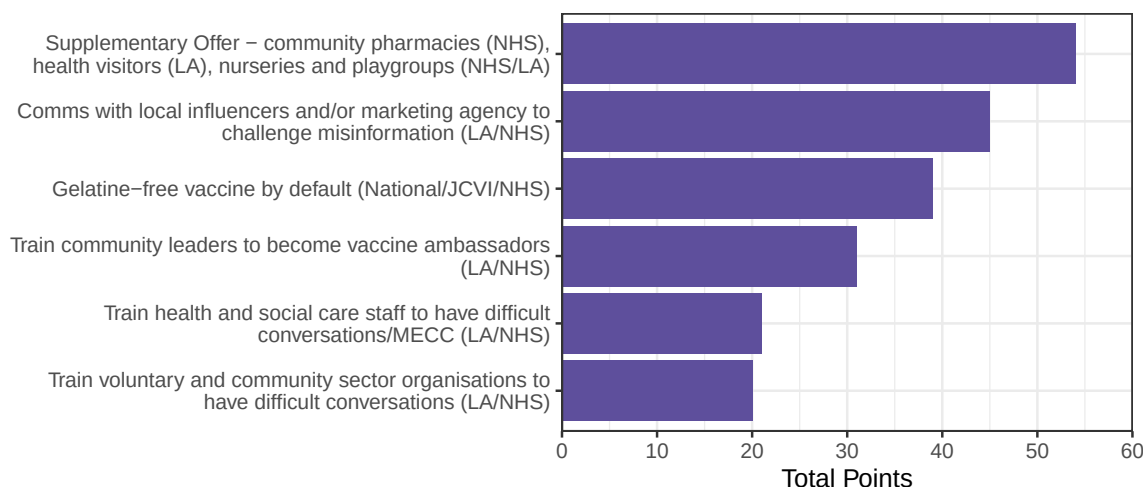


Figure 7: Anonymous participant priority ranking of interventions designed to improve measles vaccination uptake. Total score for each intervention calculated using a Borda count approach.

As shown in Figure 7, the most highly ranked intervention was an extension of the vaccination offer. This intervention, delivered jointly by the NHS and local authorities (LA), would involve expanding the avenues through which children could be vaccinated, including community pharmacies (NHS), health visitors (LA), and nurseries and playgroups (NHS/LA).

The second most highly ranked intervention was a local communications campaign to challenge misinformation and improve health literacy. Participants recommended that, while this work would be commissioned by the local authorities and/or the NHS, it would involve local influencers or marketing agencies. Doing so would help to ensure that campaigns are engaging, culturally relevant, and separate from

formal government branding. It was noted that social media content that is humorous and relatable tends to outperform formal health messaging and that this should be harnessed in countering misinformation.

The third most highly ranked intervention was to provide a gelatine-free vaccine as standard. It was agreed that this would improve uptake by reducing confusion and removing the need for families to specifically request the gelatine-free option. This intervention would be led by the national government, the Joint Committee on Vaccination and Immunisation, and the NHS.

3.3 Model Calibration

For the extended SIR model and the base model, the free model parameters were fitted to observed data from the 2023/24 outbreak by minimising the error function as outlined in Section 2.3.4. The fitted parameter values are shown in Table 2. Model-derived uncertainty ranges were calculated as the region of the parameter space in which the error function is within 10% of its global minimum.

Variable	SIR + Isolation	Base Model
Isolation Proportion, p_P	0.063 (0.058, 0.069)	0.094 (0.083, 0.105)
Isolation Delay, p_D (days)	94.7 (92.6, 96.8)	55.2 (48.6, 60.1)
"Adult" hospitalisation rate, p_H	-	0.039 (0.031, 0.045)
Time lag, τ (weeks)	-2	3

Table 2: Fitted values of the free parameters for the extended SIR and base system dynamics models. Model parameters are calculated by minimising the error function using data from the 23/24 measles outbreak in Birmingham and Solihull [23, 41].

For the SIR model, the isolation proportion was found to be 0.063 (0.058, 0.069) and the isolation delay was found to be 94.7 (92.6, 96.8). This means that, with the isolation mechanism modelled as outlined in Section 2.3.1, 94.7 days after the first infection occurred 6.3% of infected individuals were isolated each day.

The fitted isolation delays p_D should not be interpreted as estimates of the actual time taken by local health protection teams to detect or respond to cases. Instead, they represent effective model parameters capturing the aggregate effect of detection, reporting, investigation and intervention processes.

For the base model, the isolation proportion was found to be 0.094 (0.083, 0.105) and the isolation delay was found to be 55.2 (48.6, 60.1). This means that, as modelled, 55.2 days after the first infection occurred 9.4% of infected individuals were isolated each day. Additionally, the "adult" hospitalisation proportion was found to be 0.039 (0.031, 0.045). This means that an infected person aged 5 years or older has 3.9% probability of being admitted to hospital each day while infected. From

Equation 5, this means that the daily hospitalisation proportions for babies and children are 7.4% and 3.7% respectively. A two-dimensional projection of the minimum value of the error function along each axis of the three-dimensional parameter space is shown in Figure 8. Sensitivity analysis indicated that the derived free parameter values vary only moderately with changing ω (see Appendix 5).

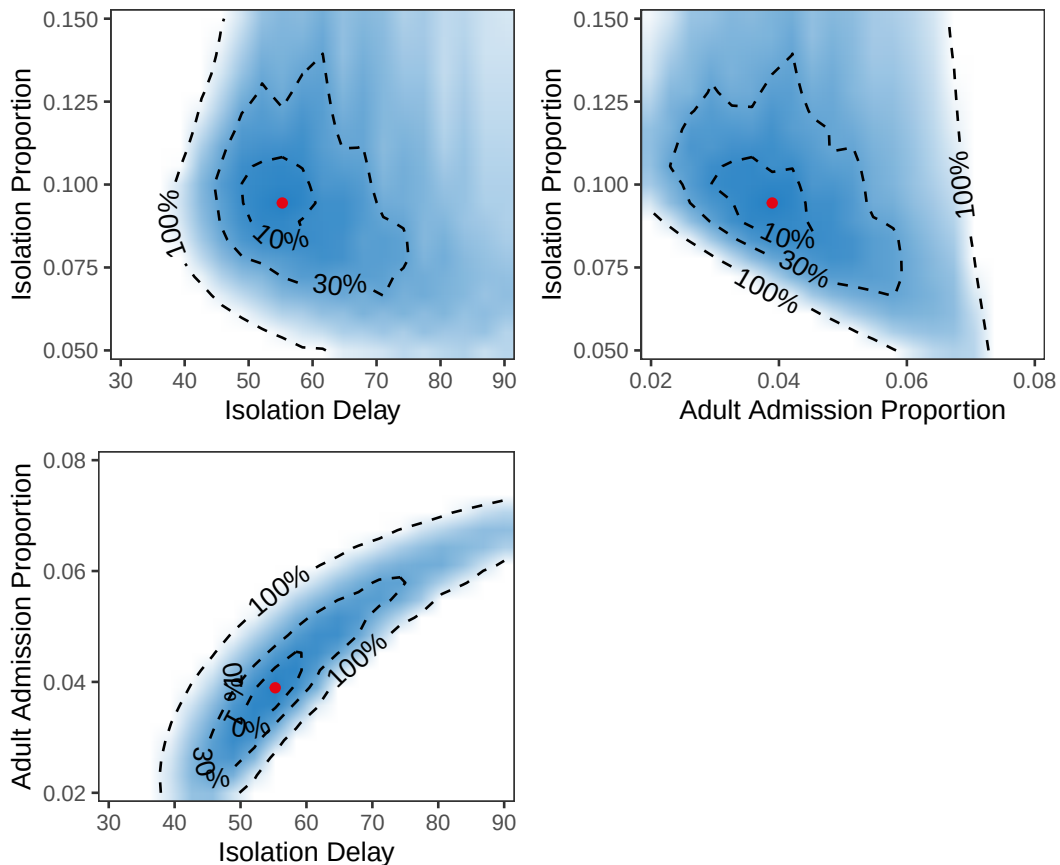


Figure 8: Two-dimensional projection of the minimum error function value along each axis of the three-dimensional parameter space for the base model presented in Figure 5. Darker shaded regions indicate where the error function is closer to the global minimum.

Figure 9 compares the epidemic curves for the three models discussed with that observed during the 23/24 outbreak. For the base model shown in Figure 5, the age-specific number of cases and hospital admissions across the outbreak are compared to the 23/24 outbreak in Table 3. The total number of modelled cases and admissions were around 92% and 98% respectively of what was observed. The distribution of infections and admissions across age was also broadly consistent with observations with a small underestimation of the number of admissions for babies.

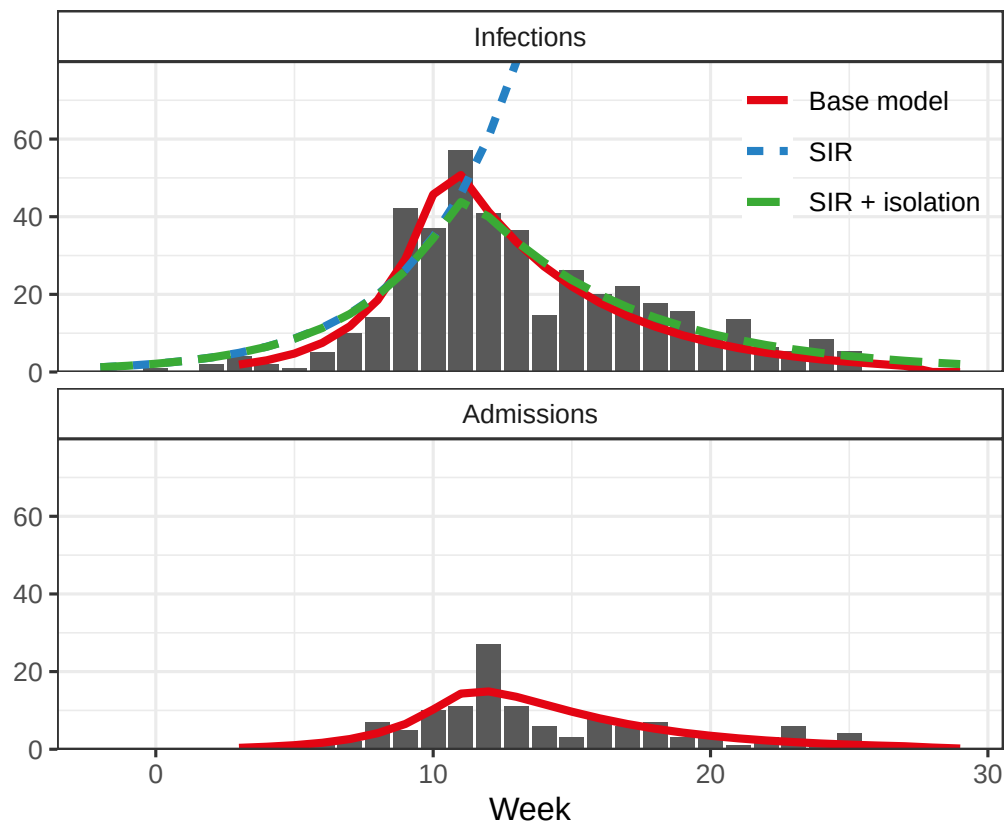


Figure 9: Weekly number of measles cases (top) and measles-related hospital admissions (bottom) during the 23/24 outbreak in Birmingham and Solihull [23, 41]. The solid red, green dashed, and blue dotted lines show the predicted epidemic curves from the base, expanded SIR, and SIR models, respectively.

Age Group (Years)	Total Cases		Hospital Admissions	
	Observed (23/24 outbreak)*	Simulated	Observed (23/24 outbreak)†	Simulated
<1	56	50.8	30	26.9
1-4	124	107.8	33	31.4
5+	226	214.4	64	66.3
Total	406	373	127	124.6

Table 3: Comparison between observed and simulated outcomes of the 23/24 measles outbreak in BSol. *Observed age-specific cases reported for Birmingham only [23]. †Calculated using SUS data from the BSol ICB data warehouse [41].

3.4 Outbreak Management

Within the presented model, outbreak management is controlled by the isolation delay p_D and isolation proportion p_P . By varying these values it is possible to estimate how different outbreak management scenarios change the estimated number of cases and hospital admissions during an outbreak.

Figure 10 shows the number of estimated cases and admissions associated with changes in the isolation delay and proportion parameters from those derived from fitting to data from the 23/24 outbreak shown in Table 2. The isolation delay was varied from 35.5 days (-36%) to 75 days (+36%), and the isolation proportion was varied from 0.0763 (-19%) to 0.150 (60%).

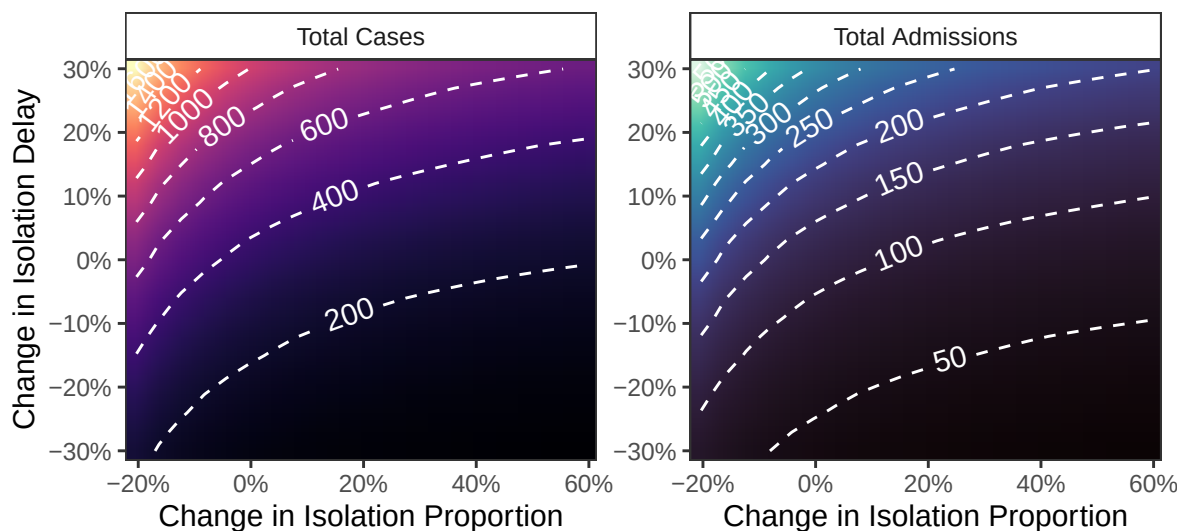


Figure 10: Estimated number of cases (left) and hospital admissions (right) with changing isolation delay and proportion parameters compared to the 23/24 outbreak.

Figure 10 shows that a small reduction in isolation proportion could greatly increase the number measles cases. For example, a 20% decrease in the daily proportion of infected individuals being isolated results in 300.3 (+83.8%) more cases and 102.6 (+83.6%) more hospital admissions. However, increasing the isolation proportion provides diminishing returns on improvements in outcomes. For example, a 20% increase in the daily proportion of infected individuals being isolated results in 88 fewer cases (-24.5%) and 32 (-24.5%) fewer hospital admissions.

In contrast, a 20% reduction in the isolation delay (11.6 days) results in 191.6 (-53.4%) fewer cases and 65.4 (-53.3%) fewer admissions. A 20% increase in the isolation delay, however, results in 378.4 (105.6%) more cases and 129.2 (105.3%) more hospital admissions.

3.5 Outbreak Preparation

Given the calibrated model, it is also possible to predict what future measles outbreaks might look like given different levels of child vaccination. Figure 11 shows the estimated number of measles infections and measles-related hospital admissions after one and five years as a function of changing relative number of MMR vaccinations. For 0% change, the number of yearly vaccinations is fixed to the average annual number of vaccines given to children aged 1 to 4 years old within BSol between 2021 and 2025.

For each annual vaccination rate value simulated, the simulation was repeated for every combination of the model parameters value including their maximum and minimum uncertainty values presented in Table 2 giving a total of $3 \times 3 \times 3 = 27$ simulations per vaccination rate. The uncertainty range shown was then determined by calculating the maximum and minimum number of cases and admissions for each vaccination rate.

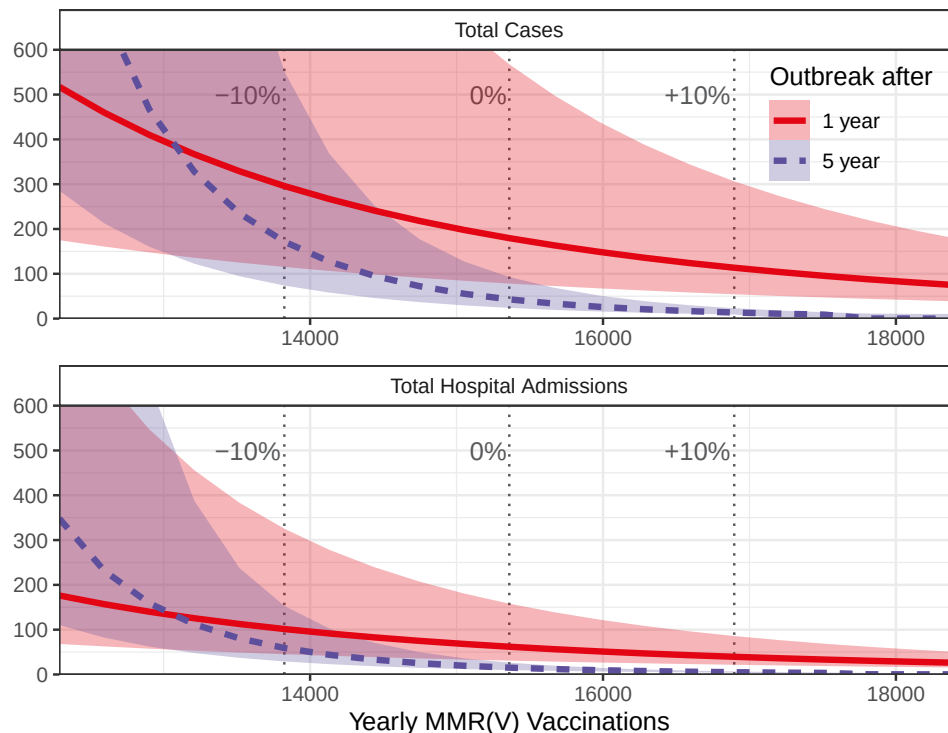


Figure 11: Total number of estimated measles cases (top) and hospital admissions (bottom) over an outbreak occurring in one (red) or five (purple) years as a function of the number of yearly child MMR vaccinations distributed.

It is estimated that, at current birth and vaccination rates, the estimated number of infections during an outbreak decreases over time. After one year, the estimated number of measles infections and hospital admissions are 179.6 (78.3, 567.7) and

61.9 (30.9, 158.1), respectively, which are 193.4 (51.8%) and 62.7 (50.3%) fewer than the simulated estimate for today (see Table 3). After five years, the estimated number of infections and hospital admissions fall to 42.9 (23.9, 92.8) and 15.2 (9.8, 26.4), respectively. Under the assumptions of the current model, the projected number of infections declines over time at current vaccination rates. This suggests that population susceptibility may decline in the coming years; however, this should not be interpreted as evidence that BSol is on a trajectory towards measles elimination due to the geographic, ethnic, and socioeconomic heterogeneity in vaccination rates.

However, Figure 11 also shows how the number of estimated infections and hospital admissions rises rapidly with falling yearly MMR vaccinations. If yearly MMR vaccinations were reduced by 10% (1,536) or 20% (3,072) then the estimated number of infections after one year are estimated to be 295.9 (115.3, 1174.9) and 517.2 (174.9, 2949.5) respectively, and after five years to be 171.8 (73.7, 551.8) and 1028.1 (285.4, 6815.9) respectively. It can be seen from the larger uncertainty ranges, that the estimates relating to lower child vaccination rates are much more dependent on the specific model parameters.

Lastly, as expected, increasing the number of yearly MMR vaccination is estimated to further reduce the number of infections and hospital admissions in the event of an outbreak. If yearly MMR vaccinations were increased by 10% (1,536) or 20% (3,072) then the estimated number of infections after one year are estimated to be 113.5 (54.2, 306.4) and 74.1 (38.3, 177.5), respectively, and after five years to be just 13.5 (9.1, 23.1) for a 10% increase. For a 20% increase, after five years, it is estimated that herd immunity is reached and no infections occur.

It should be noted that these simulations assume that the susceptible persons are mixed homogeneously amongst the population. In reality, this is known not to be the case due to the cultural, demographic, and socioeconomic trends in vaccine hesitancy and vaccine access. Therefore, even if, at a population level, the BSol area has reached the theoretical herd-immunity threshold, this may not be true for all areas across the region, meaning that outbreaks are still possible.

4 Discussion

This study presents preliminary findings from an ongoing project to develop an epidemiological model of measles transmission and hospitalisation incorporating vaccination and outbreak management in collaboration with local stakeholders. The findings include stakeholders' views on the factors perceived to influence parents' decisions to vaccinate their children, including barriers, facilitators and potential interventions to improve uptake. The development of the base model is also presented, including how changes in child vaccine uptake and outbreak management affect predicted measles cases and hospital admissions.

4.1 Participatory System Modelling

In the collaborative modelling workshop, local system experts identified the factors they believed were most important in influencing parents' willingness to vaccinate. The most highly voted individual factors were vaccine accessibility and health literacy, both with a similar number of votes (7 and 6 respectively). While "information" was the most highly voted theme (13 votes in total), accessibility received the most votes for an individual factor. This is notable because research on parents' willingness to vaccinate has focused heavily on access to information, with less attention given to other dimensions of accessibility such as availability, affordability, and proximity [65]. This is consistent with a 2024 systematic review on the barriers and enablers within the health system to childhood MMR and HPV vaccination specifically amongst disadvantaged, minority, and underserved populations in middle and high income countries found that limited time, geographic distance, and difficulties navigating the healthcare system were all factors that hindered childhood vaccination [52].

These findings contrast with much of the media and literature discourse around falling vaccination coverage which has focused on explicit vaccine scepticism [66]. Yet, a 2015 survey of 1,792 parents living in England found that only 2% of respondents reported actually refusing vaccination for their child(ren) [67]. This suggests that outright refusal may account for only a proportion of children who remain unvaccinated. Furthermore, a comparison of English vaccination coverage with parental survey data and social media sentiment similarly found that the decline in childhood vaccination coverage between 2012/13 and 2018/19 occurred alongside increasing vaccine confidence and decreasing exposure to anti-vaccination material [68].

Stakeholders also nominated and voted on interventions that they thought should be prioritised for further investigation. The most highly voted intervention was a supplementary vaccine offer through community pharmacies, health visitors, nurseries, and playgroups, followed by a communications campaign with local influencers and/or a marketing agency to challenge misinformation. These choices are broadly consistent with stakeholders' identification of accessibility and health literacy as important factors since an extended vaccine offer could directly increase accessibility and a communications campaign could help to improve health literacy and reduce the impact of misinformation.

However, offering a gelatine-free vaccine by default was voted to be the third most highly prioritised intervention for investigation despite vaccine ingredients not being identified as an important factor influencing parents' willingness to vaccinate their children against measles. The discrepancy may indicate that stakeholders' prioritisation of interventions did not simply reflect the aggregate ranking of the barriers identified in the preceding exercise.

Stakeholders also identified a potential challenge with offering a gelatine-free MMR vaccine as the default since, at the time of writing, UKHSA guidance only permits GP practices to order up to 10 doses of the gelatine-free vaccine per ac-

count per week [69]. Addressing such supply-side constraints would require action beyond individual GP practices and may therefore require consideration of national vaccine procurement and supply arrangements.

4.2 Model Fitting

The free parameters of the extended SIR model and base model were both calibrated to data from the 23/24 outbreak, as outlined in Section 2.3.4. Both models produced weekly case estimates that were in strong agreement with the observed epidemic curve. The base model also showed strong agreement with weekly hospital admissions. The base model underestimated the cumulative number of cases by around 10%. However, the distribution of cases across the three modelled age groups was broadly consistent with observations. Age-dependent hospital admissions also closely matched observations, but were slightly low for babies.

The model was calibrated and assessed against the same outbreak, meaning that its performance on independent outbreaks has not been assessed. As shown in Figure 1, the 23/24 outbreak resulted in many more cases than the three outbreaks preceding it. Potential explanations for the increased impact of the 23/24 outbreak include the large clusters of susceptible individuals in places like Birmingham following prolonged suboptimal vaccination coverage, potentially exacerbated by the disruptive impact of the COVID-19 pandemic on routine immunisations [23, 70]. Therefore, rather than providing a generic representation of a measles outbreak, the model should be considered a representation of the transmission dynamics observed in BSol during the 23/24 outbreak.

4.3 Outbreak Management

By varying the parameters of the model, estimates were made about how responding to an outbreak differently would impact the number of measles infections. The model predicted that reducing the proportion of infected people isolated each day could result in a substantial increase in infections. However, increasing the proportion isolated p_P produced only a marginal reduction in cases. Meanwhile, reducing the delay in isolation p_D was estimated to have a big impact in reducing the number of cases.

This outbreak management modelling highlights that reducing the effective delay between outbreak onset and implementation of isolation (encoded in this model via p_D) has a greater potential for suppressing the overall outbreak than improving the effectiveness of the case-finding efforts (encoded in this model via p_P). This finding is consistent with a 2021 evidence-based review of the effectiveness of contact tracing for the mitigation and suppression of viral diseases, as well as a 2021 systematic review of contact tracing methods for controlling measles outbreaks in low and middle-income countries [71, 72]. This also arises mathematically from the

exponential nature of infectious disease [73]. For these reasons, the World Health Organisation recommends that, to create a surveillance system capable of eliminating measles and rubella, the early detection of outbreaks should be given the highest priority [74].

However, it is important to note that p_D encodes all of the delays in people being isolated. For example, as well as the time it takes for health protection teams to identify that an outbreak is taking place, it also includes the time taken to identify contacts and communicate their exposure to them.

The National Measles Guidelines recommend using mass messaging such as email, text or letter [10]. During the 23/24 outbreak, Birmingham Children's Hospital Emergency Department contacted patients that had potentially been exposed to measles via letter. A retrospective study of the impact of the outbreak on the Department recommended that text messages should replace exposure letters to speed up communication [75, 76]. A future evaluation could investigate the methods used by all parts of the healthcare system to communicate exposure to highly communicable diseases such as measles. This review should then look for ways to reduce delays in communication.

Additionally, the effective time to isolation can be reduced by increasing public awareness of the symptoms of measles. Such public awareness campaigns should focus on communities with the lowest MMR uptake. Doing so has a dual benefit. Firstly it increases the probability that cases are seen by health professionals as early as possible and thus reported to the local authority (as required by the Health Protection Regulations 2010 [77]). Secondly, due to the success of vaccination programmes, vaccine preventable diseases like measles become less visible which can reduce the perceived risk from the disease [78]. Sharing materials to help parents identify measles symptoms because of the increasing outbreak frequency could help to challenge this belief.

4.4 Outbreak Preparation

Finally, the model was then used to investigate how varying levels of child vaccination might impact the total number of cases and hospital admissions of future outbreaks. It was estimated that if the number of yearly child vaccinations remained at the average for the last five years, the number of cases in future outbreaks would reduce over time. This finding, however, appears to be inconsistent with the fact that MMR coverage in BSol is well below the 95% WHO target for measles elimination and falling [22]. All else being equal, falling vaccination coverage should result in a susceptible population that grows over time meaning a greater number of infections during a subsequent outbreak. Furthermore, an additional 10% of MMR vaccination administered over the last five years were given to individuals aged 5 years or older (see Appendix 5). However, only child vaccinations have been included in the model, thus raising the level of inconsistency further.

This counterintuitive result appears to arise primarily from a mismatch between the modelled number of births and the modelled number of annual vaccinations. The model uses a single birth rate ($16,346 \text{ year}^{-1}$) determined from the 2021 ONS Census but the vaccination rate ($15,360 \text{ year}^{-1}$) is calculated based on the data from 2021 to 2025 [15, 24]. Based on these values, we would expect MMR vaccination coverage to tend towards 94% over time. In contrast, only 86.2% of the current cohort of children aged between 5 and 6 years living in BSol are recorded as having received at least one dose of MMR. A major reason that the estimated future coverage is so high is because birth rates in BSol have been falling over at least the past five years [79]. The model therefore combines a current annual birth estimate with a historical annual count of vaccination events, rather than modelling vaccination coverage within successive birth cohorts. This issue could potentially be rectified in future versions of the model by modelling vaccination as a proportion of each eligible birth cohort, rather than a fixed annual number of vaccinations.

However, irrespective of the potential overestimation of the future vaccine coverage, this analysis still provides some useful insights. The model predicts a non-linear relationship between the yearly number of child vaccinations and the estimated number of infections and hence hospital admissions during an outbreak. Increasing the number of vaccinations results in marginal reductions in infections while reduced vaccination rates result in comparatively larger increases in infections. As the outbreak occurs further into the future, this relationship becomes more non-linear. Small reductions in vaccination compound over successive birth cohorts, resulting in a progressively larger susceptible population and hence cases if an outbreak occurs. This finding is particularly important for illustrating the potential risk posed by further declines in child MMR vaccination rates.

4.5 Next Steps

In future versions of this model, once the vaccine coverage overestimation issue has been resolved, simulations of this type can be used to estimate the vaccination coverage required to prevent sustained measles transmission in BSol. This could include an analysis of the impact of increasing catch-up vaccinations for those aged five years and older who have not received a dose of MMR.

As discussed, the current transmission model assumes a homogeneous mixing of the susceptible population [80]. However, since there are strong geographic, ethnic, and socioeconomic trends in vaccination, the susceptible population is likely to be spatially and demographically clustered [23]. Some of this heterogeneity can be accounted for by stratifying the model by ethnicity. Additionally, since vaccination coverage varies substantially between ethnic groups, outbreak dynamics may depend on which communities are initially infected. Studying transmission dynamics could help health protection teams identify communities in which outbreaks are likely to have the greatest impact. These results could be used to inform the targeting of

vaccination and measles-awareness interventions towards communities at greatest risk.

Future versions of the model will also be expanded to include some of the key barriers and facilitators to MMR vaccination identified by local system experts in Figure 6. The system dynamics approach adopted allows for these factors to be represented as interacting variables within the model. However, assigning quantitative values to abstract constructs such as health literacy or misinformation presents a challenge. An additional challenge will be determining how these factors should be combined to represent parents' overall willingness to vaccinate. Estimating the current values for each of these factors and their relative influence on parents' willingness to vaccinate will require further reviews of the literature, empirical data where available, and additional stakeholder engagement.

Furthermore, the interventions most highly prioritised by stakeholders will also be implemented into the model to investigate their potential impacts on overall and ethnic inequality in vaccination and measles infections and hospital admissions. Where these interventions are expected to affect barriers to vaccination differently for different population groups, these differences can be included in the model to assess whether they reduce or widen existing inequalities in vaccination and hence infections and hospital admissions. The modelled outcomes for these interventions could provide stakeholders with evidence about which interventions are likely to have the greatest population-level impact as well as which are most effective at reducing health inequalities.

As discussed, the parameters for the model presented have been calibrated to data from the 23/24 outbreak in BSol. The external validity of the model could be assessed and potentially improved by estimating its parameters using data from a range of different outbreaks. Simulations using the resulting range of parameter estimates could then be used to quantify how variation between outbreaks affects modelled outcomes, providing an additional source of uncertainty around future projections.

4.6 Limitations

The findings of this study should be interpreted in light of several limitations.

The literature review summarised in Section 3.1 specifically focused on factors that may prevent parents from vaccinating their children and did not systematically consider factors that facilitate vaccination. As this literature was presented to stakeholders before and during the workshop, it may have biased participants to identify barriers rather than facilitators, potentially influencing the factors subsequently identified and prioritised.

As highlighted in Section 3.1, barriers to vaccination vary between population groups. Therefore, having a singular collective vote for the whole system does not capture this variation and risks producing a one-size-fits-all representation of the

determinants of vaccination uptake. A better approach might be to work with stakeholders to first segment the population into relevant groups and then identify the most important factors separately for each group.

Additionally, the process of voting on the “most important” factors may have been subject to several sources of potential bias, including the size and legibility of the handwriting on each sticky note and where they are placed. Given more time, a more comprehensive approach for eliciting stakeholders’ views on the relative importance of the identified factors would be pairwise comparison. In this, stakeholders are presented with two factors at a time and asked to choose which is most important. The individual choices can then be aggregated into an overall ordering [81]. The challenge for this method is that for a moderate number of factors, the number of comparisons can quickly become very cumbersome. A less labour-intensive approach might be the Best–Worst Scaling method. This method presents stakeholders with a series of smaller subsets of factors and asks them to identify the most and least important factor within each subset. The resulting choices could then be analysed to estimate a latent importance score for each factor, providing a more systematic basis for ranking the factors to be included within the model [82].

The stakeholder workshop also involved a small number of local system experts with a majority working at Birmingham City (7 of 10) Council. In particular, no one working in primary or secondary care was included and no one from UKHSA was able to attend. The results of the workshop therefore cannot be assumed to represent the views of all professionals working in measles prevention and control. In addition, people directly affected by measles vaccination and the interventions intended to increase uptake, including parents, schools, and members of communities with lower MMR uptake, were not directly involved in the participatory modelling. Involving participants from these groups could have provided additional insight into barriers to vaccination that may not be apparent to those included, as well as challenging professional assumptions about how parents or communities would respond to potential interventions.

There are also several limitations in the current model. For instance, the model assumes that hospitalised patients are effectively isolated and therefore do not contribute to onward measles transmission. However, during the 23/24 outbreak, Birmingham Children’s Hospital Emergency Department identified in-hospital contacts for 95 of the 161 hospital-confirmed cases included in the study, resulting in 2397 exposure letters being sent [75]. Although it is not known how many of these contacts subsequently developed clinical measles, this provides evidence that hospital settings may contribute to measles exposure and that assuming complete isolation of hospitalised cases may underestimate transmission.

Additionally, while the model considers the inward migration of individuals to BSol from abroad, it does not consider outward migration or internal migration to and from other areas of the UK. Internal migration is particularly important as it is known that more people enter BSol from other areas of the UK than abroad [83]. Future versions of the model should incorporate both internal migration into and out of BSol,

in addition to international migration.

Furthermore, the model was calibrated against the same 2023/24 outbreak against which its performance was assessed. As a result, the close agreement between modelled and observed outcomes does not demonstrate external validity, and the model's ability to reproduce independent measles outbreaks remains untested.

5 Conclusion

This dissertation presents the early development and preliminary evaluation of a system dynamics model of measles vaccination, transmission, and hospitalisation in BSol, drawing on local stakeholder knowledge and expertise to identify key barriers and facilitators to child vaccination and potential interventions to improve uptake. This model was then used to explore how varying levels of annual MMR vaccination and different outbreak management scenarios affect the estimated number of measles infections and hospital admissions.

Among the barriers and facilitators identified, stakeholders ranked vaccine accessibility and health literacy as the highest-priority individual factors for BSol, while factors under the theme of "information" received the most votes overall. Stakeholders also recommended potential interventions to improve MMR uptake and voted to prioritise extending the delivery of the vaccine into community settings and targeted communications as interventions for further investigation using system dynamics modelling.

The calibrated model reproduced the broad epidemic curve and pattern of hospital admissions with good agreement to observations. The age distributions of infections and hospital admissions were also broadly consistent, with some underestimation of the total number of cases and the number of hospital admissions for babies.

The investigation of different outbreak scenarios estimated that reducing the total effective delay between the start of an outbreak and the isolation of infected individuals could substantially reduce the number of cases and hospital admissions, while increasing the rate at which infected individuals are identified would result in more marginal reductions. Additionally, changing the yearly rate of childhood MMR vaccinations resulted in a non-linear effect on the resulting outbreak burden, with reductions in vaccination producing disproportionately larger increases in infections and hospitalisations. This effect became larger with longer periods of reduced vaccination before an outbreak.

However, the present model was calibrated and assessed against data from the same 2023/24 outbreak and therefore its external validity remains untested. It also does not represent the heterogeneous transmission dynamics associated with differences in vaccination coverage between population groups. Future development will therefore focus on validating the model against multiple outbreaks, improving the representation of population movement and childhood vaccination coverage, and stratifying the model by ethnicity to model differences in vaccination coverage and

therefore susceptibility between population groups. The behavioural factors and potential interventions identified by stakeholders will also be incorporated and evaluated.

This study illustrates the potential for a system dynamics model built in collaboration with local system experts to model population susceptibility, measles transmission, and hospital admission. Following further development, including incorporating the behavioural factors and potential interventions identified by stakeholders, such a model could provide a valuable decision-support tool for intervention planning and outbreak response.

Ethical Approval

Ethical approval was not required for this study. The data were routinely collected for health service evaluation and were provided in either pseudonymised or fully anonymised form. Additionally, participants involved in this study were acting in their professional capacities as representatives of partner organisations. The NHS Health Research Authority's self-assessment tool indicated that NHS Research Ethics Committee review was not required.

Generative AI statement

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A Vaccine Hesitancy Literature Table

Title	Authors	Year	Citation
"The shadow of reactogenicity: Parental hesitancy due to prior vaccine experiences"	Gschwend et al.	2026	[26]
"Religious barriers to measles vaccination"	Wombwell et al.	2015	[53]
<i>SMBC Community Engagement Learning</i>	Solihull Metropolitan Borough Council	2025	[42]
<i>UASC Health & Wellbeing Survey</i>	Kemishi	2025	[43]
"Parental vaccine hesitancy: Recent evidences support the need to implement targeted communication strategies"	Cagnotta et al.	2025	[25]
"Vaccine-preventable diseases in migrants in Europe: a systematic review"	Halliday et al.	2025	[58]
"Parental vaccine refusal, non-vaccinated children, and outbreaks of Vaccine-Preventable Diseases (VPDs) in Europe: a systematic review of aetiology and risk"	Baiocchi et al.	2025	[84]
<i>Report of Community Engagement Workshops in Birmingham: Feedback on UKHSA Published Materials on Measles</i>	Palmer et al.	2024	[44]
<i>Summary of Birmingham Measles Soft Intelligence</i>	Bissett et al.	2024	[45]
"Understanding the health system barriers and enablers to childhood MMR and HPV vaccination among disadvantaged, minority or underserved populations in middle-and high-income countries: a systematic review"	Essa-Hadad et al.	2024	[52]
"Prevalence of caregiver hesitancy for vaccinations in children and its associated factors: A systematic review and meta-analysis"	Maamor et al.	2024	[85]

Table 4: Publications on vaccine hesitancy as outlined in Sec 3.1.

Title	Authors	Year	Citation
“Why parents say no to having their children vaccinated against measles: a systematic review of the social determinants of parental perceptions on MMR vaccine hesitancy”	Novilla et al.	2023	[28]
“Vaccine hesitancy in migrant communities: a rapid review of latest evidence”	Tankwanchi et al.	2021	[86]
“MMR vaccine attitude and uptake research in the United Kingdom: a critical review”	Torracinta et al.	2021	[27]
“Resurgence of measles in Europe: a systematic review on parental attitudes and beliefs of measles vaccine”	Wilder-Smith et al.	2020	[61]
“Parents’ and informal caregivers’ views and experiences of communication about routine childhood vaccination: a synthesis of qualitative evidence”	Ames et al.	2017	[87]
“Socioeconomic differences in childhood vaccination in developed countries: a systematic review of quantitative studies”	Bocquier et al.	2017	[46]

Table 4: Publications on vaccine hesitancy as outlined in Sec 3.1.

B Transcribed Models

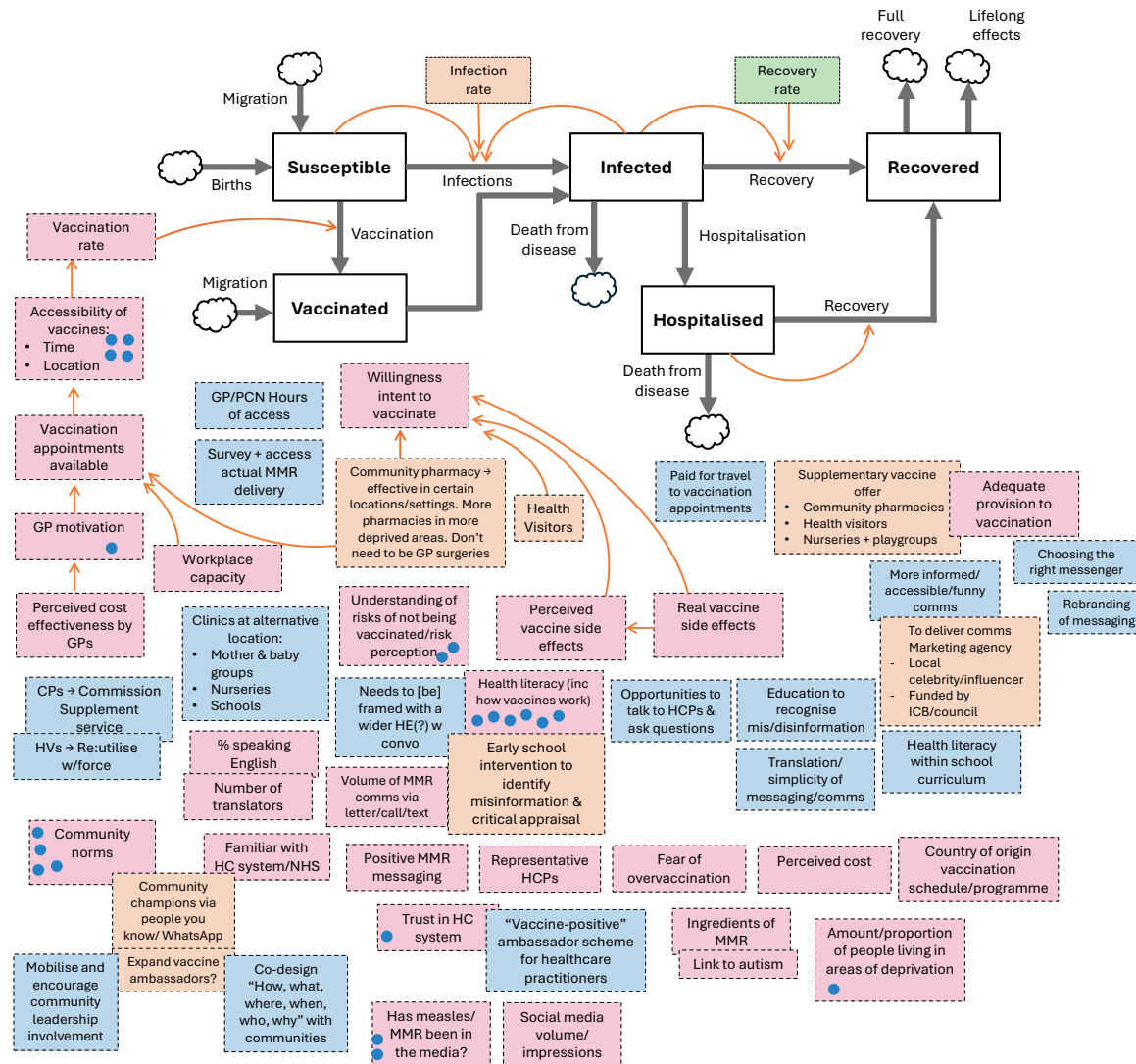


Figure 12: Factors influencing population flows and potential interventions for the BSol measles vaccination, infection, and hospitalisation system. Digitally transcribed from the notes taken by Group One at the stakeholder system modelling workshop.

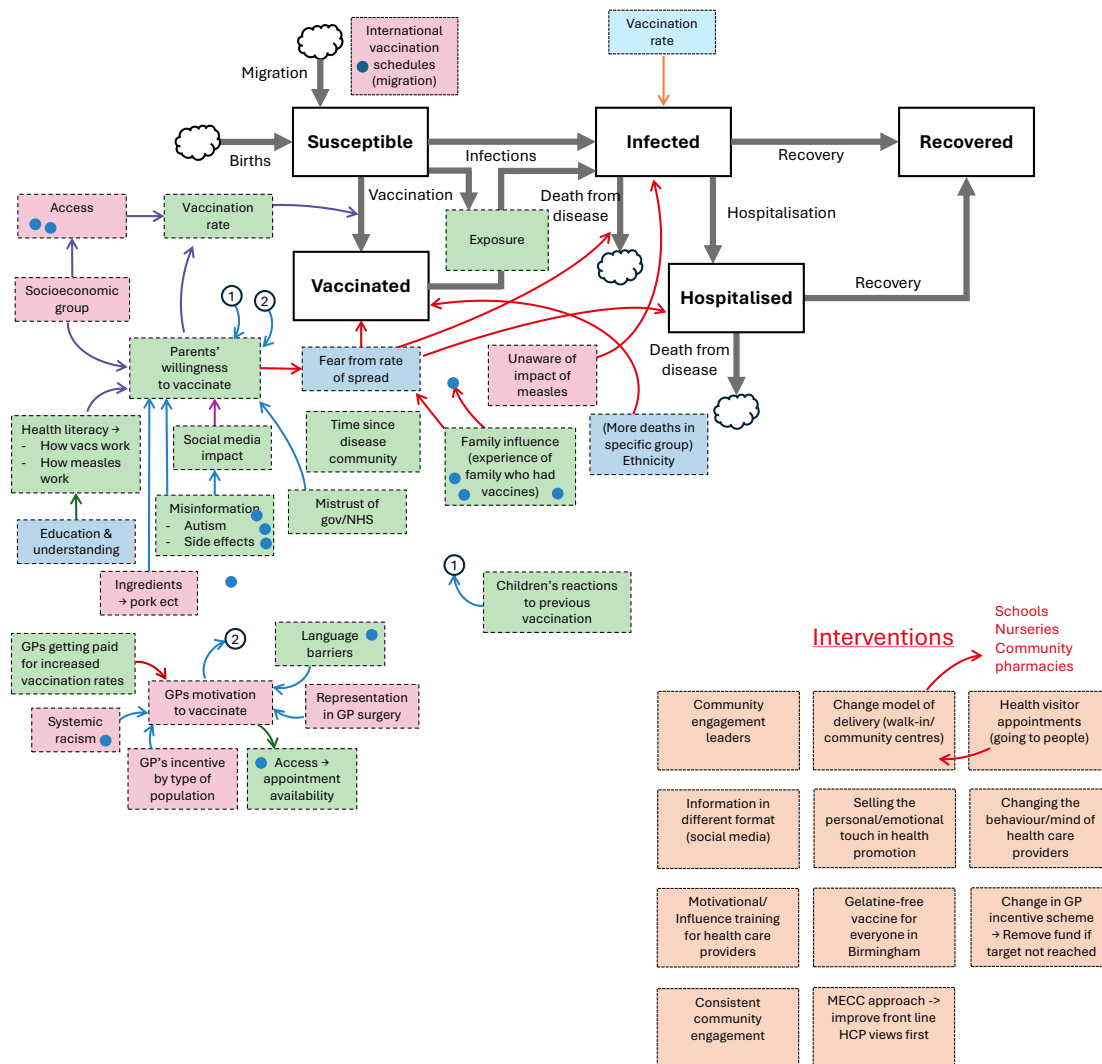


Figure 13: Factors influencing population flows and potential interventions for the BSol measles vaccination, infection, and hospitalisation system. Digitally transcribed from the notes taken by Group Two at the stakeholder system modelling workshop.

C Calculating Observed Outcomes Data

The number of weekly measles cases in Birmingham and the number of weekly hospital admissions (ICD10 code “B05”) in Birmingham and Solihull from 13/10/2023 to 12/04/2024 are shown in Figure 14. Case data for Birmingham were extracted from Jary et al., who defined a case to be “a Birmingham resident with laboratory confirmation (measles IgM in blood or oral fluid in the absence of recent vaccination (within 14 days) or confirmed wildtype measles RNA in any clinical specimen)” [23]. This definition differs from that used in UKHSA’s public measles reports, which only include lab-confirmed cases. This can be seen from the fact that Jary et al. report 175 Birmingham cases in 2023 starting from October 13th, while UKHSA report only 118 for all of 2023 [88].

Weekly case data for Solihull was not available. For 2023, annual data for Solihull was not reported since there were fewer than ten cases. For 2024, UKHSA reported a total of 364 and 16 lab-confirmed cases for Birmingham and Solihull, respectively. Due to Solihull’s geographical proximity to Birmingham, it is likely that the majority of the cases for the local authority occurred at the same time. If the weekly cases reported for Birmingham by Jary et al. is C_B , then we can approximate the weekly cases in Solihull to be

$$C_S = \frac{16C_B}{364}. \quad (8)$$

This approximation simultaneously accounts for the reduced reporting period and the additional cases not confirmed by a lab reported by Jary et al. While this is a crude approximation, since Solihull contributed 4.2% of the UKHSA-reported cases in 2024, it is unlikely that this will have a significant impact on the fitting of the model to these data.

D Model Parameters

Vaccination data was extracted from the NHS Federated Data Platform (FDP) [24]. The yearly number of vaccinations in BSol from 2021 to 2025 are presented in Figure 15. It can be seen that there was a substantial increase in vaccinations to those aged five to 25 years in 2024 due to the “catch up” vaccination program conducted during the 23/24 measles outbreak [23]. The annual child (one to four years) rate was taken to be the average number of yearly vaccines administered to children over this five years.

A relatively crude method was used to estimate the number of vaccinated and unvaccinated persons entering BSol from abroad. Firstly, the 2021 census was used to calculate the number of babies (aged < 1 year) and children (aged 1-4 years) living in BSol from each major world region: Africa, Antarctica and Oceania, Europe (not UK), Middle East and Asia, The Americas and Caribbean [15, 39]. The counted babies had only one year to enter the country, while the children had four years.

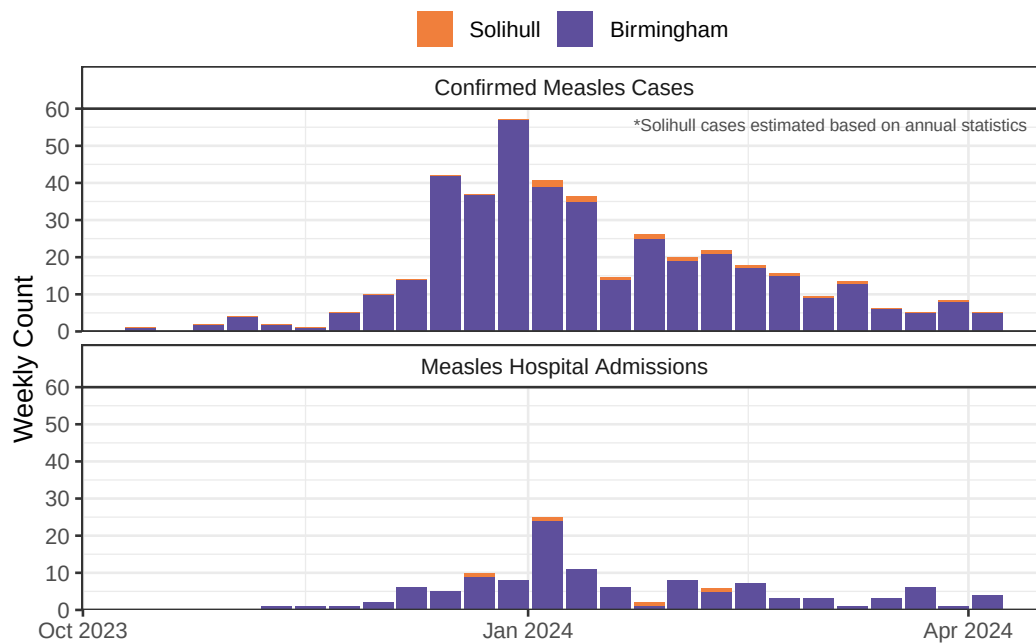


Figure 14: Lab-confirmed measles cases (top) and measles-related hospital admissions (bottom) in Birmingham and Solihull from 13/10/2023 to 12/04/2024. Birmingham case data extracted from Jary et al., with Solihull data approximated from UKHSA annual data [23, 88]. Hospital SUS data extracted from the BSol ICB data warehouse.

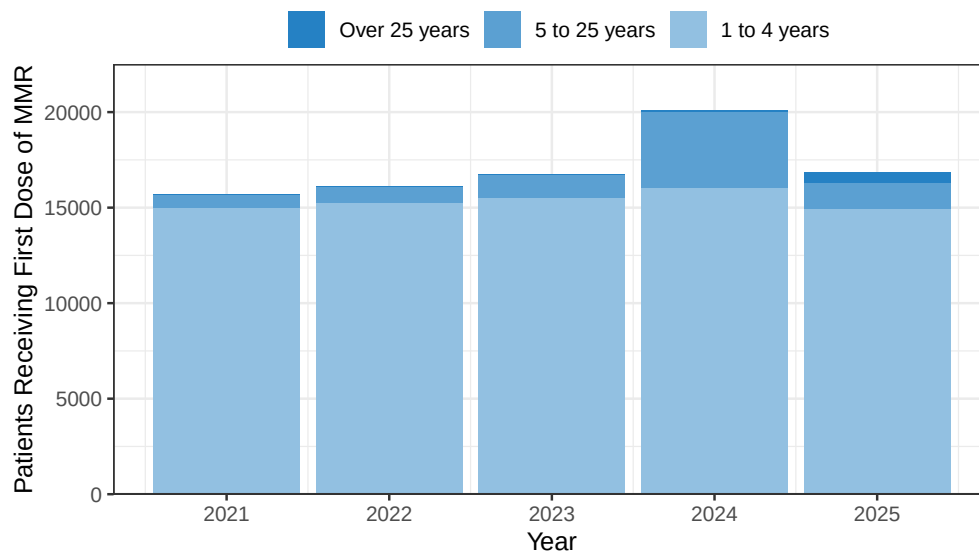


Figure 15: Number of first doses of MMR or MMRV administered by BSol ICB each year from 2021 to 2025 [24].

Therefore, The yearly migration number for babies was therefore taken to equal the census count of babies, while the yearly migration number for children was taken to be the census count divided by four.

It was then assumed that all babies were unvaccinated and that children were vaccinated at the same proportion as the WHO estimates for the percentage of the population from each region that have received at least one dose of measles-containing vaccine [40]. The region-specific estimates are shown in Figure 16. From this, it is estimated that there are 226 babies and 882 children entering BSol from abroad each year, with 90.9% of the children having received at least one measles-containing vaccine.

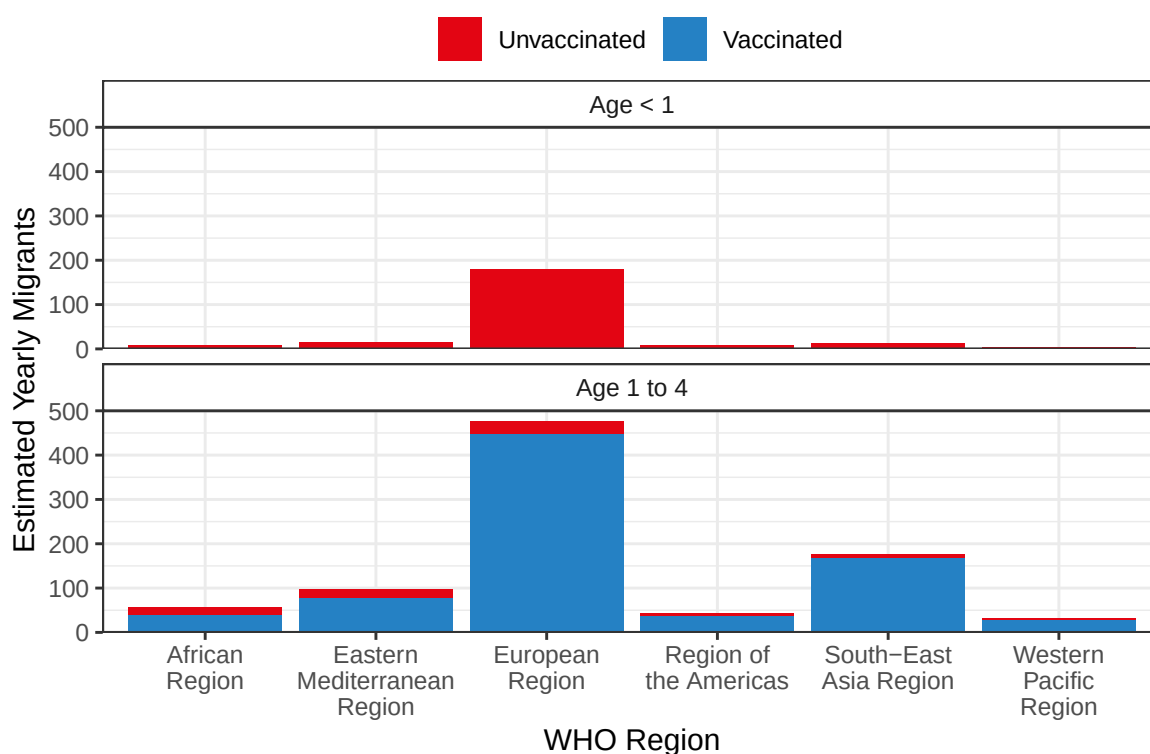


Figure 16: Estimated number of vaccinated (blue) and unvaccinated (red) babies (top) and children (bottom) migrating each year to BSol from each of the WHO regions [15, 39, 40].

The above estimate for child migration assumes that children migrated into the country at a consistent rate over the four years preceding the census. While this is a reasonable approximation over a four-year time scale, it is inappropriate to make the same assumption for all adults. Therefore, the "adult" migration rate was estimated by comparing the number of people aged five years or more who were born outside of the UK living in BSol in the 2011 and 2021 censuses [15, 39]. From this, it is estimated that 6305.2 adults migrate into BSol each year. It is then assumed that

the current pattern of migration origins for adults matches that of children and that therefore 90.9% of these adults are vaccinated.

These estimates should reflect the rough ball-park figures for migration into BSol. However, importantly, they do not account for migration out of BSol or for internal migration to and from other areas within the UK. The latter is the most significant omission, as it is known that more people enter BSol from other areas of the UK than abroad [83]. Future models should take this additional incoming population source into account.

A full list of parameters used in the base system dynamics model is presented in Table 5.

Variable	Value	Source(s)
Initial number of babies (<1 yrs)	16,346	[15]
Initial number of children vaccinated (1 to 4 yrs)	69,365	[15, 24]
Initial number of children unvaccinated (1 to 4 yrs)	14,620	[15, 24]
Birth rate	16,346 yr ⁻¹	[15]
Baby (<1 yrs) migration rate	196 yr ⁻¹	[15]
Vaccinated child (1 to 4 yrs) migration rate	801.7 yr ⁻¹	[15, 40]
Unvaccinated child (1 to 4 yrs) migration rate	80.1 yr ⁻¹	[15, 40]
Vaccinated "adult" (5+ yrs) migration rate	5731.4 yr ⁻¹	[15, 40]
Unvaccinated "adult" (5+ yrs) migration rate	573.8 yr ⁻¹	[15, 40]
MMR/MMRV vaccination rate (1 to 4 yrs)	15,360 yr ⁻¹	[24]
Measles basic reproduction number, R_0	13.5	[89]
Measles infectious period, $1/\gamma$	1.25 weeks	[90]
Median hospital stay	2 days	Locally calculated [41]

Table 5: Model parameter values used in the measles system model presented in Figure 5.

E Initial Conditions

Table 6 presents the initial numbers in each population state used when calibrating the base system dynamics model and investigating different outbreak response scenarios. When investigating outbreak preparation through varying vaccination rates, the initial number of infected people was set to zero, with a single infected adult introduced after a fixed period of time.

State	Babies (<1 year)	Children (1-4 years)	"Adults" (5+ years)
Susceptible	16,346	14,620	136,364
Infected	0	0	1
Isolated	0	0	0
Vaccinated/Recovered	0	69,365	1,138,068
Hospitalised	0	0	0

Table 6: Initial conditions used when calibrating the base model presented in Figure 5.

F Sensitivity Analysis

Figure 17 shows that the free parameters p_D , p_P , and p_H , vary only marginally with changes in ω . Importantly, at this resolution, the parameters are the same for $\omega = 0.5$ and for $\omega = 1$.

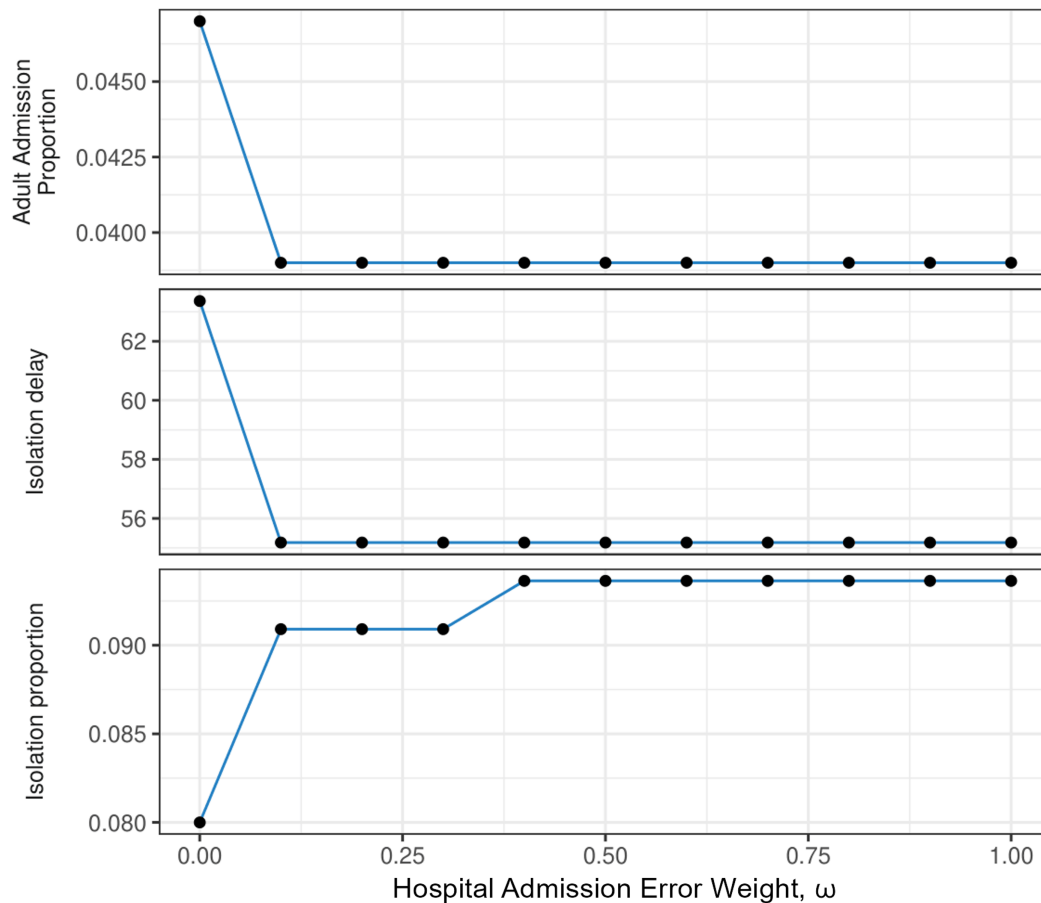


Figure 17: Variation in the derived values for free model parameters adult admission proportion p_H (top), isolation delay p_D (middle), and isolation proportion p_P as a function of the relative weight of the hospital data in the error function calculation ω .